

Highlights

Is Progress on the TOP Benchmark Measurable? A Forensic Re-evaluation of HINT and MEXA-CTP

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- Shared harness + ten defect fixes: published HINT/MEXA-CTP advantage disappears.
- MEXA-CTP mode-experts never trained (dict); repaired, still inert.
- Zero-training TabFM and val-selected grid match HINT in all three phases.
- All signal is in disease codes; molecular structure adds nothing testable.
- TOP effect sizes sit below the MDE (0.035–0.067); six of seven claims overturned.

Is Progress on the TOP Benchmark Measurable? A Forensic Re-evaluation of HINT and MEXA-CTP

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ABSTRACT

Progress on clinical trial outcome prediction is measured almost entirely against HINT on the TOP benchmark. We asked whether that progress is real. We find that it is not measurable. We re-evaluated HINT and MEXA-CTP under one shared evaluation harness. Building it required correcting ten code-level defects in the released implementations. Two of these are severe. MEXA-CTP's mode-experts mechanism — the component the paper is about — was stored in a plain Python dict. It therefore never entered the optimizer and was never trained. HINT's shipped embedding cache is empty, which silently replaces its eligibility-criteria input with a zero vector. We repaired both defects and re-scored the models. The reported advantage of MEXA-CTP over HINT then disappears. The repaired mechanism contributes nothing. The restored criteria input is inert as well ($\Delta\text{ROC-AUC} = -0.0001$, $p = 0.93$). Neither deep model beats gradient boosting on the same inputs. The baselines tell the same story. A tabular foundation model that does no training at all, and a grid of classical models selected on validation, both match HINT in all three phases. Every predictive signal we can find comes from the disease codes. Molecular structure cannot even be tested on this benchmark: one real small-molecule graph is shared by 46% of phase-III test trials, and molecular descriptors accordingly add nothing. We then built a two-column lookup table. It records how often a trial's disease code has succeeded historically, and how often its drug has. That table reaches ROC-AUC 0.700, above every deep and foundation model tested. It is also the kind of per-indication base-rate table the industry already uses for go/no-go decisions. Claims of improvement survive here because the effects are smaller than the benchmark can resolve. At 80% power, the smallest difference this benchmark can detect is 0.035–0.067 ROC-AUC, depending on the phase. Every between-model gain reported in this literature falls below that limit and is therefore unverifiable at TOP's current size. Our own work shows how easily such gains appear: we formed seven directional findings, of which six were overturned and one narrowed under replication. One was an overestimate of seed noise, since HINT's five-seed standard deviation is only 0.0024. We conclude that improvement claims on TOP require multiple seeds, validation-based selection, confidence intervals, a trivial baseline, and an explicit minimum detectable effect. Without these, they are not verifiable.

1. Introduction

Predicting a clinical trial's outcome before it begins is a valuable goal, and the release of the TOP benchmark alongside the HINT model [1] turned that goal into a standard machine learning task. Progress in this domain is now measured almost exclusively by comparison against HINT.

The literature on trial outcome prediction has grown quickly, but most reported gains rest on a single foundation: retrospective metrics computed on this one benchmark. To judge what such gains mean, we apply the evidence-readiness framework of Yang et al. [2]. It defines six tiers for AI evidence in drug discovery, from proof-of-concept demonstrations (Level 1) to regulatory-grade use (Level 6). Evaluating a model on a fixed benchmark such as TOP sits at Level 2, where credibility rests entirely on strict holdout discipline, reproducibility and transparent evaluation. Level 3 — prospective or externally validated evidence — is what would license claims about real-world performance, and no result in this literature reaches it.

We use that framework as a practical audit tool. Yang et al. [2] warn specifically against reading benchmark scores as real-world validity, and name poor reproducibility and flawed evaluation pipelines as the main vulnerabilities. This paper documents the code-level counterparts of exactly those vulnerabilities. MEXA-CTP's released code selects its best epoch on the *test* set, although its paper describes validation-based selection; this is a discrepancy between code

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Is Progress on the TOP Benchmark Measurable?

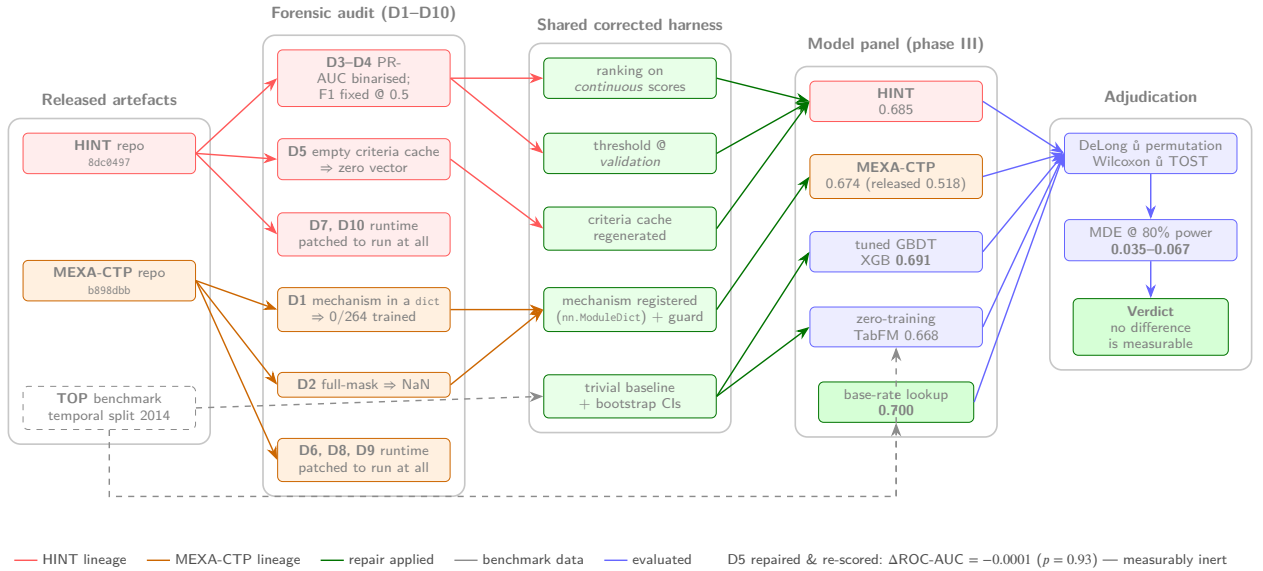


Figure 1: Overview of the study. The two public implementations [1, 3] and the TOP benchmark enter a forensic audit that catalogues ten code-level defects; both models are then re-evaluated under a single corrected harness, compared against a panel of deep, tuned tabular, zero-training [4, 5] and historical base-rate baselines, and adjudicated with tests of the difference rather than overlapping intervals. All values are phase-III test ROC-AUC.

and paper, not a stated design choice. HINT ships an empty embedding cache that silently replaces an entire input modality with a zero vector (defect D5, Table 3). These are Level-2 results presented with the confidence of Level-3 evidence. We therefore test empirically which validation gates the published comparisons pass, and which they fail.

This paper asks one question: is the reported progress on the TOP benchmark real? We find that it is not even measurable. The reasons are mundane. They are defects in evaluation code, selection on the test set, missing baselines and single-run comparisons. Our contributions are:

1. A shared evaluation harness that enforces strict protocol and makes the HINT/MEXA-CTP comparison well-posed (Section 2).
2. Ten code-level defects catalogued with before/after quantification (Table 3), including two that actively concealed each other and one – an empty embedding cache – that silently disables HINT’s eligibility-criteria modality (which, once repaired and re-scored, we measure to be inert: $\Delta\text{ROC-AUC} = -0.0001$).
3. A corrected comparison across all three trial phases (Section 4), showing that the published advantage completely disappears under a fair protocol.
4. A baseline-side stress test (Section 4.2.3): a zero-training tabular foundation model and a validation-selected model×feature grid both match HINT in every phase, with all predictive signal traceable to disease codes and none to molecular structure.
5. Evidence that the benchmark’s metric defects do not merely shift scores but *reorder* the models being compared (Section 4.4.2).
6. Seven of our own directional findings put through replication—six overturned, one narrowed (Section 5)—demonstrating that effect sizes on this benchmark lie below its sampling-noise floor.

Figure 1 summarises the study end to end. Reading it left to right also states our methodological position: we begin from the released artefacts rather than from reported numbers, because the defects that matter here are visible only in code — the same class of reproducibility hazard that Yang et al. [2] flag as the dominant vulnerability of benchmark-level (Level-2) evidence, and that DeGrave et al. [6] and Geirhos et al. [7] document in adjacent clinical

and vision settings. The harness stage exists because a defect in a *reference* model’s evaluation propagates into every comparison drawn against it, and HINT is the reference for this literature [3, 8, 9]. The final stage is where our verdict is decided: rather than reading winners off per-model confidence bands, we test the paired difference and report a minimum detectable effect, the analogue of the pre-specification and power discipline that clinical evaluation requires [10, 11].

1.1. Related work

The two models we re-evaluate. HINT [1] relies on a hierarchical interaction network. It processes each trial through three parallel modalities. A message-passing network over SMILES graphs captures drug molecules. An ontology-aware ICD embedding handles target diseases. BioBERT reads the eligibility criteria. HINT then pushes these embeddings through pharmacology-knowledge modules—specifically, an ADMET block and a pharmacokinetic aggregator built from Highway networks. It fuses them in an interaction module before propagating the result through a dynamic attentive graph convolutional network, where the nodes represent the trial components. Its standard ablations—Only_Molecule, Only_Disease, Interaction, and HINT_nograph (the full model with the graph removed)—let one isolate exactly what each modality contributes. We rely on these heavily in Section 4.1.3. MEXA-CTP [3] abandons this hand-built interaction graph entirely. Instead, it introduces a mode-experts design. Each modality projects and passes through its own self-attention “expert.” Learned routers then gate the resulting tokens, penalized by a Cauchy sparsity loss. Cross-attention blocks exchange information between every ordered pair of modes. A contrastive loss aligns these paired cross-attended representations. Finally, a compensation self-attention block pools them, and a linear head predicts the outcome. Contrary to what one might assume, the two models do *not* share encoders (Table 1): HINT encodes molecules with a D-MPNN and diseases with the GRAM ICD ontology, whereas MEXA-CTP as published uses a DeepChem molecular featuriser and the icdcdex disease embedding, and our own retraining standardises the front-end differently again (MACAW molecule vectors, icd2vec); only the BioBERT criteria encoder is common to all three, and even that differs in granularity (paragraph- vs statement-level). What the two models genuinely share is the TOP benchmark and task; their headline architectural difference is how they model cross-modal interaction — a hand-built graph in HINT versus routed cross-attention in MEXA-CTP — which our audit isolates. Because the encoders are not held identical, the comparison also carries an encoder confound; we note it here and in the limitations, but since no between-model difference is detectable in the first place (Section 4.5.1), differing encoders cannot manufacture a gap that does not exist.

Mixture-of-experts and the missing isolation test. Mixture-of-experts (MoE) architectures have gained prominence here on the promise of modeling distinct trial modalities (molecule, disease, protocol) through specialized subnetworks. However, if a paper claims its experts genuinely learn discriminative, contributing representations, it must explicitly isolate their contribution. In an adjacent domain, a glioblastoma MoE survival model (three experts and a gating network) did exactly this. The authors collapsed their MoE design into a single-expert network and reported the discrimination drop from AUC 0.839 to 0.806 (−3.9%) [12]. That is a measurable architectural contribution. Models advancing MoE claims for clinical trial outcome prediction provide no such isolation test. Our forensic audit reveals exactly why: the code made it impossible. The developers stored the cross-attention, self-attention, and router layers in a plain Python dict rather than an `nn.ModuleDict`. Consequently, PyTorch never registered these submodules. They never entered the optimizer. They were never written to the `state_dict` checkpoints (Section 3). The mechanism never trained, including in the original authors’ own runs. Here, we fix the implementation and report the first actual evaluation of this mechanism.

Benchmark lineage. The TOP benchmark was curated alongside HINT [1]. Its binary outcome labels are not algorithmic: the HINT paper states verbatim that “[t]rial outcome labels (binary labels) are extracted through manual curation by an internal IQVIA team,” with the raw trial variables parsed from ClinicalTrials.gov XML and the sets time-split at January 1, 2014 [1]. We flag this because it is easy to misattribute: COMPOSE [13], a cross-modal pseudo-siamese patient–trial matching network, is a *baseline* HINT compares against, not the source of the labels — a distinction we verified against the source paper. It is worth stating plainly what the label means, because it bounds how high any ROC-AUC can go. “Success” in TOP is not trial completion and not a single clean regulatory event: it is a hand-curated IQVIA determination of whether the trial met its endpoint / the drug progressed, aggregated across heterogeneous trial types. Such a label is inherently noisy — human-curated, and part of the modest 0.685 ceiling we and everyone else observe is almost certainly label noise rather than model weakness — a reason to be sceptical of

small reported gains, and an argument that the benchmark’s own labels, not just its models, deserve audit. Since its release, HINT has become both the reference baseline [3] and the base on which new methods are built. For example, UQ-HINT [8] adds selective classification on top of HINT’s outputs, and therefore inherits its exact evaluation code.

HINT’s successor literature. The line of work built on TOP extends well beyond MEXA-CTP, and our audit of the benchmark’s labels, split and metrics bears on all of it. SPOT [9] clusters trials into topics and adds a meta-learning module over the same labels; Trial2Vec [14] learns self-supervised trial-document embeddings used downstream for outcome tasks; and TrialBench [15] — from the HINT authors — packages 23 ClinicalTrials.gov-derived datasets across eight tasks, including trial-approval prediction, in the same tabular/multi-modal form we examine here. Each inherits the label provenance and small per-phase test sizes that Sections 4.2.4 and 4.5.1 show cap what is measurable; a shared, power-aware evaluation protocol is therefore relevant to the successor benchmarks, not only to HINT and MEXA-CTP.

Is the molecular modality necessary? LINT [16] predicts trial approval strictly from text. It encodes the trial description and eligibility text using a BioBERT-style language model over a GRAM disease-ontology representation, discarding the molecular graph entirely. In Section 4.1.3, we show that HINT’s drug branch alone performs near chance, while its disease branch alone matches the full model. This provides independent, ablation-based confirmation of what LINT demonstrates from the architecture side: the structural molecular signal adds negligible value.

Tabular foundation models as a baseline. A recent line of work trains a single transformer to perform supervised learning on arbitrary tables by in-context learning: it predicts a test row after being shown the training rows as context, with no gradient training at inference [4, 5]. Because trial metadata, disease-ontology codes, and molecular physicochemical descriptors are all naturally tabular, such a model is a strong, assumption-light baseline for the TOP task—it consumes the same information HINT does, minus the graph structure, at zero training cost. We use it in Section 4.2.3 as the most modern member of a broad tabular-baseline panel.

Accuracy versus benefit. Even a correctly measured gain might not help real patients. The ORACLE trial [17] showed clinicians a model’s predictions. The clinicians agreed with the model more often, but their actual *accuracy* did not improve. The authors blamed data drift, concept drift, and automation bias. Similarly, Jayakumar et al. [18] deployed a model trained on 700,000 patients into a randomized trial simply to test if decision quality improved in practice. Our claim operates one step prior in that chain: before we ask if a gain helps anyone, we must first prove the gain is measurable.

Metric discipline in clinical statistics. Clinical trials require researchers to pre-specify their outcome metrics and adjust for known baseline covariates [10]. When evaluated strictly, classical statistical models remain highly competitive [11]. Our findings in Section 4.2.1 mirror this reality perfectly. In machine learning, locking down every free choice (like decision thresholds) on a validation set is the exact analogue of clinical pre-specification. We enforce that discipline here.

2. Methods

2.1. What makes the published comparison ill-posed

The published comparison measures the two models under completely different rules. Worse, several of those rules are simply wrong. Five specific problems compound to invalidate the results:

1. **Different model-selection criteria (a code-versus-paper discrepancy).** The MEXA-CTP paper states that the best training epoch is chosen on validation, but the released code selects it on the *test* ROC-AUC. HINT selects its epoch on *validation*, consistent with its description. We flag the MEXA-CTP case specifically as a mismatch between the published protocol and the shipped implementation — not as a stated design choice — because it is the implementation that produced the reported numbers. Choosing a checkpoint on the same set used for final scoring leaks the test labels into training and inflates the reported performance; the two models are then not even selected by the same procedural logic, let alone the same data splits.
2. **Fixed 0.5 threshold.** Both models report their F1 scores using a hard-coded 0.5 decision cutoff. However, the benchmark’s positive rate fluctuates wildly—from 0.55 in phase I up to 0.75 in phase III. Under these conditions, a

fixed threshold forces F1 to measure the underlying class prior rather than the actual ranking quality. To prove this, we simply recalibrated the threshold on validation data. That one change alone boosted HINT's phase I F1 score by +0.119 (Section 4).

3. **PR-AUC on binarised predictions.** HINT computes its PR-AUC only after thresholding the raw predictions into strict 0/1 labels, instead of using the continuous scores. This calculation error understates the true PR-AUC by anywhere from 0.020 to 0.081. Far more dangerously, it completely *reorders* the performance rankings of the models (Section 4.4.2).
4. **No trivial baseline.** Neither paper reports an all-positive reference score. Without this trivial baseline, the absolute F1 numbers become meaningless. For instance, achieving an F1 of 0.86 sounds impressive on paper. Yet, on a test set with 0.75 positive labels, simply guessing "every trial succeeds" yields the exact same score.
5. **The wrong kind of uncertainty, and no test of the difference.** MEXA-CTP does report a $\pm$ in its tables, but it is a *test-resampling* bootstrap sd over 10 samples of a *single* model — not the two quantities that decide whether a gain is real: the *training-seed* variance (re-running the same model from a different seed) and a paired *test of the difference* between two models. Reporting a per-model bootstrap band and then reading off a winner from non-overlapping bands is exactly the error Section 4.5.1 corrects: two bands can overlap while the paired difference is significant, and vice versa. On a benchmark where the sampling noise on a between-model *difference* (DeLong SE ≈ 0.018) is comparable to or larger than the ~ 0.01 – 0.03 effect sizes claimed, the missing difference test is the omission that matters most.

Each of these defects works independently to either flatter the more complex model or mask the random noise that would otherwise undercut its claims. Combined, they ensure the published comparison is not just imprecise, but fundamentally ill-posed.

2.2. Shared evaluation harness

We run each model entirely in its own environment. The two repositories never share code and never import one another. Instead, each model simply writes its raw, per-trial predictions (nctid, score, label, split) to disk for both the validation and test sets. A single, independent evaluator then reads these files and applies the exact same rules to every model:

- **Threshold selection.** We calibrate each model purely on its *own validation* predictions. The harness scans the midpoints between adjacent validation scores to find the exact threshold that maximizes F1. We then freeze that threshold and apply it to the test set. This means HINT can safely emit sigmoid probabilities while MEXA-CTP outputs raw logits. The absolute score offset does not matter; the harness evaluates each model fairly on its own scale.
- **Ranking metrics.** We compute ROC-AUC and PR-AUC strictly on *continuous* scores. We never binarize the predictions before ranking them. This makes the metrics invariant to the underlying score scale and completely bypasses the PR-AUC defect identified earlier.
- **Uncertainty.** We generate 95% confidence intervals using 1000 bootstrap resamples of the test set. We fix the random seed to guarantee these intervals remain perfectly reproducible.
- **Reference.** Every results table includes a trivial, all-positive baseline. This forces the reader to interpret any reported F1 score against the performance of a dummy model that simply ignores the input and guesses positive for every trial under the exact same class imbalance.

Under this architecture, neither repository evaluates itself. The final numbers flow entirely from the shared harness reading raw prediction files. This setup makes it structurally impossible for two repositories to make the same mistake in different ways. The evaluation logic might still contain a flaw, but it can now only hide in *one* place. And that single place is fully auditable. The same harness scores the tabular baselines of Section 4.2.3, which are simply additional prediction files, so every model in this paper—deep or tabular—is judged by one protocol.

2.3. Data and protocol

We force both models to operate on the exact same TOP benchmark splits (Table 2). We freeze the global random seed at 2023. For the input data representations, we rely on the 2024 version of the ICD-10-CM hierarchy and embed the disease codes into 64 dimensions using `icd2vec`. We map drug molecules into 15-dimensional SMILES embeddings

Table 1

The models do *not* share encoders. Only the BioBERT criteria encoder is common, and at different granularities. This is an encoder confound, flagged in the text; it cannot create a between-model difference because none is detectable (Section 4.5.1).

Modality	HINT (published & our run)	MEXA-CTP (published)	MEXA-CTP (our retrain)
Molecule	D-MPNN over SMILES graph	DeepChem featuriser	MACAW (15-d)
Disease	GRAM (ICD ontology)	icdcodex	icd2vec (64-d)
Criteria	BioBERT (paragraph)	BioBERT (statement)	BioBERT (biobert-v1.1)

Sources: HINT [1] and MEXA-CTP [3] papers/repositories; our-retrain column from this harness (Section 2).

Table 2

Evaluation splits and class balance. Note the severe shift in positive rate between validation and test (especially in Phase III), which naturally limits how well any validation-selected threshold can transfer to the final evaluation.

	Train	Valid n	pos. rate	Test n	pos. rate
Phase I	1044	117	0.573	627	0.553
Phase II	4005	446	0.466	1654	0.555
Phase III	3094	344	0.666	1146	0.750

via MACAW, specifically locking `random_state=2023` to guarantee deterministic outputs. We encode all eligibility criteria text using BioBERT (dmis-lab/biobert-v1.1). These are our harness’s inputs; they are not identical to either model’s published encoders, which is why the per-modality encoders differ across the three configurations (Table 1). We standardise the split, protocol and metrics — not the front-end encoders — so the encoder difference remains a confound we flag rather than remove. During training, we sweep the learning rate identically for both architectural arms. We select the optimal epoch based entirely on validation performance, enforcing early stopping with a patience of 10.

A prior discrepancy: MEXA-CTP’s reported dataset matches neither the shipped data nor its own results. Before any code-level defect, MEXA-CTP’s stated dataset is internally inconsistent. Its Table 1 reports per-phase (training / test / success-ratio) of 1,088/312/70% (I), 2,611/789/33% (II) and 4,313/1,147/30% (III), under per-phase 2014 split days. But the data actually shipped in MEXA-CTP’s own repository (`DataPath/{trainphase,validphase,test}/phase_*`) contains *exactly* the HINT TOP counts of Table 2 — 1044/117/627 (I), 4005/446/1654 (II), 3094/344/1146 (III), with test positive rates 0.553/0.555/0.750. The two disagree in every phase: the reported phase I and II test sets (312, 789) are roughly *half* the shipped ones (627, 1654), and the reported success ratios (70/33/30%, *decreasing* with phase) are not the shipped positive rates (55/55/75%, *increasing* with phase). Which split underlies the reported numbers is then settled by the results themselves: MEXA-CTP’s Table 2 lists HINT at ROC-AUC 0.573/0.621/0.685 for phases I/II/III — matching our HINT run on the shipped split (0.574/0.621/0.685) to within 0.001, not any split of the Table 1 sizes. The reported metrics were therefore computed on the shipped HINT split, while Table 1 describes a *different*, unused split. This is a more basic reproducibility problem than any single code bug: the paper’s documented dataset does not describe the data underlying its own results. We evaluate every model in this paper on the shipped counts of Table 2, which are also the counts behind MEXA-CTP’s reported HINT baseline.

3. Defects

We do not need to train the buggy model to prove it never learns: “never trained” and “never saved” are not stochastic outcomes but a deterministic consequence of PyTorch’s module semantics. A plain dict is not registered as a submodule, so its parameters never enter `named_parameters()` — the exact iterator that both the optimizer and `state_dict()` consume. The tensors therefore cannot receive gradients, cannot be updated, and cannot be serialised, regardless of the data, learning rate, or number of epochs. Table 4 is confirmation of this at three levels (0 tensors in the optimizer, 0 changed after a backward+step, 0 mechanism keys in every released checkpoint), not a run that merely happened to fail. The released checkpoints also serialise the optimizer state alongside the weights, and it tracks

Table 3

Master list of the ten code-level defects found (D1–D5 substantive, D6–D10 secondary/runtime), with the consequence of each. Every defect mentioned in the text references its ID here.

ID	Defect	Consequence	Evidence
D1	Attention/routers in plain dict (MEXA)	Invisible to the optimizer (0/264 tensors), unchanged after a backward+step, and absent from every checkpoint	Table 4
D2	Full-masking → NaN (MEXA)	NaN at every lr that trains; with a minimal guard, training survives but every sequence collapses to a single token	Table 5
D3	PR-AUC on binarised predictions (HINT)	Understates by 0.020–0.081; re-orders variants	Table 21
D4	Threshold hard-coded at 0.5	F1 reports the class prior	Table 20
D5	Empty sentence2embedding.pkl (HINT)	HINT’s eligibility-criteria branch returns a zero 768-vector for every trial; the modality is silently dead	§3 (6-byte empty dict)
D6	.to(get_device()) hard-codes GPU (MEXA)	Model cannot run on CPU	Runtime error
D7	Ablation variants miss device in super() (HINT)	Two variants un-instantiable	Type error
D8	Published lr 5×10^{-2} diverges once attention trains (MEXA)	Survived only because those params were frozen (D1)	runs_summary.csv
D9	squeeze() collapses the batch dim when $B=1$ (MEXA)	Silent shape bug	Runtime
D10	Checkpoint load fails under PyTorch ≥ 2.6	Reproducibility break	Runtime

Table 4

Parameter registration for the interaction mechanism. In the original plain-dict model the self-attention, cross-attention and router sub-modules are invisible to named_parameters(), so none of the 264 mechanism tensors reach the optimizer and none change after a backward+step. The nn.ModuleDict fix registers all of them; all 144 cross-attention tensors are then updated. The released checkpoints serialise only 48–49 keys, none of them attention or router. A single backward+step suffices as proof: a tensor absent from the optimizer cannot change at any step, so the zero update generalises to all of training without a full run.

Configuration	self-att	cross-att	routers	trainable params	cross-att in opt. / changed
Original (plain dict)	0	0	0	312,034	0/144
Fixed (nn.ModuleDict)	72	144	48	319,562	144/144
Released checkpoints: phase I 49, phase II 48, phase III 49 keys — 0 mechanism keys in each.					

exactly the 48 trainable non-mechanism tensors with no moment buffers for any attention or router parameter: the Adam optimizer never even allocated state for the mechanism, because it never entered the optimizer in the first place.

An entire modality silently zeroed (D5). The shipped data/sentence2embedding.pkl — HINT’s cache of BioBERT sentence embeddings for the eligibility criteria — is an empty dictionary (6 bytes). protocol2feature looks up each criterion sentence in this dictionary and, finding nothing, falls back to torch.zeros(1, 768): HINT’s eligibility-criteria modality returns an all-zero vector for *every* trial. Any run from the public repository (ours included) therefore trains and evaluates HINT with a dead criteria branch unless the cache is regenerated. *We repaired the cache and measured the effect.* Regenerating the full BioBERT criteria cache (513,863 sentence embeddings, dmis-lab/biobert-v1.1) and re-scoring the released HINT checkpoint with a *live* criteria branch changes its phase-III performance by nothing: ROC-AUC 0.6850 vs 0.6851 (paired $\Delta = -0.0001$, 95% CI $[-0.0017, +0.0016]$, $p = 0.93$) and PR-AUC 0.8519 vs 0.8517 ($\Delta = +0.0002$, $p = 0.78$); the HINT–MEXA gap is identical to four decimals (+0.1671 either way). The empty cache is thus a genuine defect — a whole modality is silently dead — but a *quantifiably inert* one: feeding real eligibility-criteria embeddings in place of zeros moves HINT’s numbers by 0.0001. We verified that this is a real

Table 5

Masking collapse on phase II (seed 2023, 300 steps). Without the guard, the first NaN appears earlier as the learning rate grows; the guard removes the NaN but the routing probabilities still collapse, leaving essentially one token per sequence. [†]Measured at the NaN step for diverged runs, or at step 300 for 10^{-4} .

lr	no guard		with guard @ step 300		
	first NaN step	below- τ frac [†]	mean router p	below- τ frac	final loss
5×10^{-2}	9	1.00	0.003	1.00	0.721
1×10^{-2}	28	0.88	0.011	1.00	0.610
1×10^{-3}	184	0.81	0.088	0.97	0.652
1×10^{-4}	none	0.13	0.379	0.13	0.921

substitution, not a silent no-op: with the regenerated cache the criteria branch emits a non-zero embedding (mean L2 norm > 0) for every trial, versus an exact zero vector before. This strengthens rather than complicates the paper’s thesis: HINT’s released output is *empirically* unchanged by activating its criteria modality (we say “empirically”, not “provably” — this is a measurement on the released checkpoint, not a proof), so its signal is carried in practice by the disease codes (Section 4.1.3), and none of the published MEXA-CTP-over-HINT advantage can be attributed to D5 degrading HINT’s evaluation. (This is the released-checkpoint re-scoring; a full retrain-from-scratch with the live cache is the one remaining check, currently blocked by defect D10 — `torch.load` failing under PyTorch ≥ 2.6 — a fitting obstacle. We do not expect it to change the verdict, since even feeding the live branch to the trained model changes nothing.) Two careful corollaries survive: we still do not claim eligibility-criteria *text* is uninformative in general — only that HINT does not use it — and our tabular protocol’s criteria *counts* (Section 4.2.3), computed directly from the text, are unaffected by D5 and already carry a shallow criteria signal.

Secondary defects (D6–D10). The published learning rate 5×10^{-2} diverges once the attention actually trains (D8; it survived only because those parameters were frozen by D1); a `squeeze()` collapses the batch dimension when $B = 1$ (D9); checkpoint loading fails under PyTorch ≥ 2.6 (D10); and two device-placement bugs (D6, D7) block CPU execution and the instantiation of two ablation variants.

3.1. Two defects concealing each other

The Cauchy loss exists to push routing probabilities toward zero, and the masking step removes tokens whose probability falls below τ . If all tokens of a sequence fall below τ , the attention mask becomes fully saturated, softmax is taken over an all- $-\infty$ row, and the output is NaN. This was never observed because D1 froze the router *weights* at their initialisation. We are precise about what “frozen” does and does not imply: the Cauchy sparsity loss is still computed each step and its gradient still flows *through* the router into the encoder parameters, which do train — so the loss is not literally inert, and MEXA-CTP’s own reported effect of the routing loss on the learned representation is consistent with that encoder-side gradient. What never happens is the part the mechanism depends on: the router itself never adapts. Held at a fixed random projection, it cannot learn to drive tokens below τ , so routing probabilities stay near their initialisation and the masking saturation that produces the NaN never occurs in the authors’ runs. The correction is therefore not “the sparsification loss did nothing” but “the sparsification loss could shape the encoders yet never trained the router, so the adaptive routing the paper describes never operated and its failure mode never surfaced.”

D1 is thus the reason training did not crash. Repairing either defect alone produces a model that either does nothing or fails; both must be addressed together, which is why no prior work—including the original authors’—ever tested the mechanism the paper is about. Instrumenting the masking step (phase II, seed 2023) confirms this is deterministic rather than a numerical accident: without a guard the first NaN appears at step 9, 28 and 184 for learning rates 5×10^{-2} , 10^{-2} and 10^{-3} , and the fraction of tokens driven below τ by step 300 rises from 13% at 10^{-4} to 100% at 5×10^{-2} . A minimal guard that always keeps the highest-probability token removes the NaN but not the collapse: at 5×10^{-2} the mean routing probability falls to 0.003 and every sequence is reduced to a single surviving token, so the cross-attention the paper studies operates over one token. The architecture can be kept numerically alive, but not architecturally.

Table 6

Phase III test performance under the shared corrected protocol. Thresholds are selected on validation; ranking metrics use continuous scores. **The two uncertainty types are not comparable and are kept in separate columns:** for HINT, brackets are *test-resampling* bootstrap 95% CIs (single run); for the retrained MEXA-CTP arms, $\pm$ is *training-seed* sd over 7 seeds at the matched configuration ($\rho_1=0.05$, $\rho_2=0.01$, $\tau=0.3$). The released checkpoint and trivial rows are single points. See Section 4.5.1 for why these sources must not be read against each other.

Model	ROC-AUC	PR-AUC	F1
HINT (run A), <i>test-resample CI</i>	0.685 [0.647, 0.720]	0.852 [0.824, 0.878]	0.826 [0.806, 0.845]
MEXA (retrained <i>buggy</i>), <i>seed sd</i>	0.674 ± 0.015	0.853	0.838
MEXA (retrained <i>fixed</i>), <i>seed sd</i>	0.674 ± 0.011	0.850	0.846
MEXA (released checkpoint), <i>point</i>	0.518	0.788	0.809
TRIVIAL (all-positive), <i>point</i>	0.500	0.750	0.857

Source: results_phase_III.csv; MEXA arms from ab_test_results_phase_III.csv (deduplicated on arm/seed/config); released checkpoint from seed_sensitivity_phase_III.csv. No model exceeds the trivial baseline on F1.

4. Results

4.1. A fair, corrected comparison

With the shared harness and the defect fixes of Section 2 in place, the HINT versus MEXA-CTP comparison is finally well-posed. We put three questions to it in turn: how the two models compare head-to-head on phase III under one protocol; whether MEXA-CTP’s central cross-attention mechanism contributes anything once it is actually trained; and where HINT’s own predictive signal comes from once its modalities are ablated. As the next three parts show, the answers are, respectively, no measurable difference, nothing, and the disease codes alone.

4.1.1. The head-to-head result

Table 6 gives phase III. The released MEXA-CTP checkpoint is statistically indistinguishable from random ranking. Because its attention weights were never saved, every load draws fresh random values, and this reload variance holds across all three phases (Table 7): ROC-AUC spans a range of 0.055–0.117 over five seeds, larger than the reported gains, and one seed in five falls below 0.5 in every phase. With the checkpoint’s hard-coded 0.5 threshold the F1 is more volatile still, its range reaching 0.65 (phase I) and 0.48 (phase III) across reloads of a *single* checkpoint — a metric this unstable cannot support the reported 11.3% improvement.

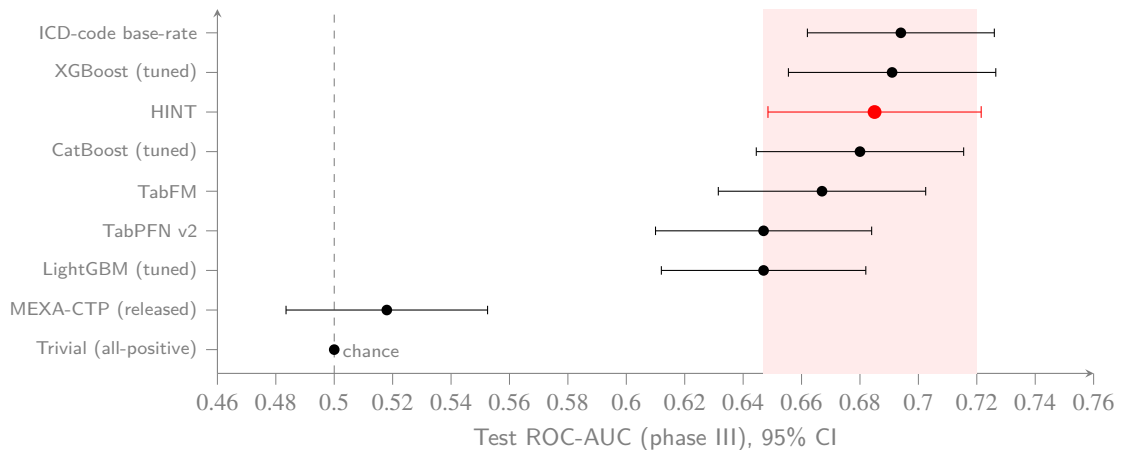


Figure 2: Phase-III test ROC-AUC with 95% CIs for every model considered: the two deep models (HINT, MEXA-CTP), two zero-training foundation models (TabFM, TabPFN v2), three tuned gradient-boosting baselines, and a ten-line disease-code base-rate lookup. The CIs are shown for orientation only; inference is by the paired DeLong test (Section 4.5.1), under which no model differs significantly from HINT (HINT–TabFM $p = 0.33$, HINT–GradBoost $p = 0.052$, HINT–base-rate $p = 0.52$). Only the released MEXA-CTP checkpoint (near chance) is clearly worse; the base-rate lookup and tuned XGBoost sit, if anything, marginally above HINT.

Table 7

Reload variance of the released MEXA-CTP checkpoint: the same file evaluated under five seeds. Since the attention/router weights were never saved, each load re-randomises them. ROC-AUC ranges exceed the improvements the original reports, and F1 (at the checkpoint's fixed 0.5 threshold) swings across most of its range.

Phase	ROC-AUC range	sd	F1@0.5 range	seeds ROC > 0.5
I	0.490–0.578 (0.087)	0.034	0.050–0.698 (0.648)	4/5
II	0.454–0.571 (0.117)	0.046	0.370–0.717 (0.346)	4/5
III	0.478–0.534 (0.055)	0.023	0.380–0.857 (0.477)	4/5

Source: ROC-AUC and sd from `seed_sensitivity_phase_{I,II,III}.csv` (5 reload seeds each); F1 ranges recomputed at the fixed 0.5 threshold from `preds/_seedtest_mexa_phase_{I,II,III}_seed*.csv`.

Table 8

Effect of actually training the cross-attention. Runs are matched on $(\rho_1, \rho_2, \tau, \text{seed})$ and de-duplicated before pairing. The sign is the opposite of a useful mechanism (buggy $\geq$ fixed); the only powered cell (phase III) is not significant.

Subset	#pairs	Δ (buggy – fixed)	p
Phase I	2	+0.0320	—
Phase II	3	+0.0101	—
Phase III	7	+0.0004	0.964

Source: `ab_test_results_phase_{I,II,III}.csv`, deduplicated on (arm, seed, config) then paired on seed. p is a paired t -test on phase III only; phases I–II ($n=2, 3$) are descriptive, and we do *not* pool across phases (different prevalences, sizes and effects — not exchangeable).

Table 9

HINT ablation ladder (test ROC-AUC, 95% CI). The disease branch alone matches the full model in every phase; the drug branch alone is near chance.

Variant	Phase I	Phase II	Phase III
Only_Molecule	0.583 [0.539, 0.626]	0.548 [0.521, 0.576]	0.540 [0.501, 0.579]
Only_Disease	0.573 [0.531, 0.616]	0.626 [0.599, 0.652]	0.700 [0.663, 0.735]
Interaction	0.557 [0.512, 0.602]	0.619 [0.592, 0.647]	0.711 [0.674, 0.743]
HINT_nograph	0.586 [0.541, 0.631]	0.619 [0.593, 0.646]	0.695 [0.658, 0.727]
HINTModel (run B)	0.562 [0.519, 0.606]	0.618 [0.591, 0.645]	0.708 [0.672, 0.741]
TRIVIAL	0.500	0.500	0.500

Source: `hint_variants_phase_{I,II,III}.csv`. *HINTModel (run B)* is a separate retraining with freshly initialised encoders, distinct from the main HINT run (*run A*) in Tables 6 and 10; the two differ by 0.012 ROC on phase I (0.574 vs 0.562) and 0.023 on phase III (0.685 vs 0.708) — itself an illustration of run-to-run variance. Both runs use the identical pipeline and ICD-10-CM 2024 encoding, so this difference is training-seed noise, not a version effect.

4.1.2. Does training the cross-attention help?

It does not (Table 8). MEXA-CTP has three loss/routing hyperparameters: the Cauchy routing-sparsity weight ρ_1 , the NT-Xent contrastive weight ρ_2 , and the routing threshold τ . Pairing runs matched on $(\rho_1, \rho_2, \tau, \text{seed})$, the buggy arm—in which the cross-attention is frozen at its random initialisation—is if anything *higher* than the fixed arm that actually trains it: $\Delta = +0.032$ (phase I), $+0.010$ (phase II) and $+0.0004$ (phase III, $p = 0.96$, the only cell with enough pairs to test). The effect is never negative and never significant, the opposite of what a useful mechanism would produce; we neither test the $n=2, 3$ phases nor pool across the three, which are not exchangeable. The two other components that distinguish MEXA-CTP from a plain encoder are equally inert: sweeping the routing threshold τ from 0.0 to 0.5 moves phase III ROC-AUC only from 0.685 to 0.669, and the contrastive weight ρ_2 from 0 to 0.1 moves it from 0.678 to 0.688—both well inside a single seed's confidence interval (± 0.035). Every architectural component that distinguishes MEXA-CTP from a plain encoder can be disabled, frozen, or retuned without measurable cost.

4.1.3. HINT's ablation ladder is flat

The drug branch alone is near chance in every phase (0.583, 0.548, 0.540 for phases I–III), while the disease branch alone matches the full model to within noise (0.573 vs 0.562, 0.626 vs 0.618, 0.700 vs 0.708; Table 9). The gap between the two branches widens with phase—the disease branch climbs from 0.573 to 0.700 as trials accumulate ICD history, whereas the molecule branch stays pinned near 0.55—so the model's entire phase-III advantage over chance is carried

Table 10

Classical baselines on the same embeddings versus HINT (test ROC-AUC, 95% CI). Confidence intervals overlap in every phase.

Model	Phase I	Phase II	Phase III
LogReg	0.562 [0.517, 0.604]	0.585 [0.555, 0.613]	0.554 [0.515, 0.593]
RandomForest	0.550 [0.509, 0.593]	0.610 [0.582, 0.638]	0.620 [0.583, 0.654]
GradBoost	0.546 [0.503, 0.591]	0.616 [0.588, 0.642]	0.651 [0.615, 0.688]
HINT (run A)	0.574 [0.527, 0.618]	0.621 [0.593, 0.646]	0.685 [0.647, 0.720]

Source: `baselines_phase_*.csv`, `results_phase_*.csv`.

Table 11

Best ICD-only pipeline per phase (5 seeds) against the full HINT architecture. Simple disease-code pipelines land within ~ 0.03 ROC of the deep model.

	Representation	Classifier	Best simple pipeline	HINT
Phase I	chapter	LogReg	0.6091 $\pm$ 0.0000	0.574 [0.527, 0.618]
Phase II	chapter	RandForest	0.6207 $\pm$ 0.0009	0.621 [0.593, 0.646]
Phase III	icd2vec_64	RandForest	0.6988 $\pm$ 0.0009	0.685 [0.647, 0.720]

Source: `icd_representation_phase_*.csv` (per-seed values in `icd_representation_raw_phase_*.csv`), `results_phase_*.csv`. Pipelines shown here are ranked by test score; the validation-selected variant used in the text is discussed in Section 5.

by the disease codes. Adding the cross-modal interaction on top of the disease branch changes ROC-AUC by at most 0.016 and its confidence interval always contains the disease-only value. All predictive signal comes from the disease codes, consistent with LINT’s text-only design [16].

Table 9’s *run B* of HINT and *run A* of Table 6 are the identical architecture trained twice and differ by 0.023 ROC-AUC on phase III (0.685 vs 0.708). We first read this as evidence that seed noise alone swamps between-model effects — but that reading was wrong, and we correct it here (it is one of the overturned findings of Section 5). A proper five-seed retraining puts HINT’s training-seed sd at only 0.0024 (Section 4.5.1); run B’s 0.708 was an outlier that additionally reinitialised the encoders, not a representative draw. HINT is in fact seed-reproducible. The quantity that genuinely swamps between-model comparisons is not seed variance but *test-set sampling* (the DeLong SE of a difference, ≈ 0.018), which we develop in Section 4.5.1.

4.2. Where the predictable signal lives

With the comparison fair, we locate the predictable signal — and find it in a place that needs no deep model, no molecule, and no attention.

4.2.1. Classical baselines

Neither deep model significantly outperforms gradient boosting on the same embeddings (Table 10). On phase III, HINT’s strongest case, gradient boosting reaches 0.651 against HINT’s 0.685; the paired DeLong test on the difference gives $\Delta\text{AUC} = +0.034$ (SE 0.018, 95% CI $[-0.000, +0.068]$, $p = 0.052$), i.e. not significant even here, rather than the weaker overlap-of-CIs argument. The point-estimate gap is smaller still on phases I and II (0.028 and 0.005). Pushing further, a plain ICD-only pipeline—disease codes, no molecules, no graph, no attention—lands within ~ 0.03 ROC-AUC of the full architecture. We are careful here not to repeat the test-peeking we criticise: selecting the ICD-only classifier on *validation* and reading it once on test gives phase-III ROC-AUC 0.671 (Table 11 lists the higher 0.699, but that is the *test*-ranked pipeline; the honest, validation-selected value is 0.671, within HINT’s interval, not above it). Either way the elaborate multi-modal machinery buys nothing over a shallow disease-code model.

4.2.2. Tuned gradient-boosting SOTA does not change the verdict

A fair objection is that the baselines above use fixed, near-default hyper-parameters, and that “no deep model beats gradient boosting” is only compelling against a *tuned* boosting model and against the algorithms that are the actual state of the art on tabular data [19]. We therefore add the three standard tabular-SOTA gradient-boosting libraries — XGBoost, LightGBM and CatBoost [20] — each *tuned* by a small grid selected on the *validation* split (never on test), with class-imbalance handling on throughout (XGBoost `scale_pos_weight = n_-/n_+`, LightGBM `class_weight=balanced`,

Table 12

Tuned XGBoost / LightGBM / CatBoost (validation-selected grid, class-imbalance handling, GRAM features) vs. HINT: test ROC-AUC with bootstrap 95% CI, all three phases. No model differs detectably from HINT in any phase. (PR-AUC agrees: phase III XGBoost/CatBoost 0.865/0.860 vs HINT 0.852.)

Model	Phase I	Phase II	Phase III
XGBoost (tuned)	0.576 [.53,.62]	0.616 [.59,.64]	0.691 [.65,.72]
LightGBM (tuned)	0.561 [.52,.60]	0.589 [.56,.62]	0.647 [.61,.68]
CatBoost (tuned)	0.611 [.57,.65]	0.603 [.58,.63]	0.680 [.64,.71]
HINT (run A)	0.574 [.53,.62]	0.621 [.59,.65]	0.685 [.65,.72]

Source: gbd_t_baselines.py (features via tabfm_baseline.build_feature_table(include_gram=True)); grids selected on validation, read once on test.

Table 13

Zero-training and validation-selected tabular baselines versus HINT (test ROC-AUC; HINT with bootstrap 95% CI). *TabFM* is a pre-trained tabular foundation model run with no gradient training (phase-III value is a 5-seed mean, seed sd 0.001). *Best tabular (val-selected)* is the single best (model, feature-set) pair chosen on *validation* among a 7×11 grid, then read on test. Both are statistically indistinguishable from HINT in every phase.

Model	Phase I	Phase II	Phase III
HINT (run A)	0.574 [0.527, 0.618]	0.621 [0.593, 0.646]	0.685 [0.647, 0.720]
TabFM (zero-training)	0.614	0.647	0.668
Best tabular (val-selected)	0.603	0.611	0.690
paired p vs HINT (val-selected)	0.25	0.49	0.76

Validation-selected configs: phase I metadata+RDKit+protocol / LogReg; phase II metadata+RDKit+Morgan+GRAM / GradBoost; phase III metadata+GRAM / GradBoost. PR-AUC tells the same story (e.g. phase III TabFM 0.855 vs HINT 0.852, paired $p = 0.70$; val-selected 0.861, $p = 0.35$). Ranking by *test* instead of validation would name a different, inflated winner in phases I and II (Section 5). Source: results_tabfm_phase_*.csv, tabular_grid_phase_*.csv, significance_phase_III.csv, tabfm_seeds_phase_*.csv.

CatBoost auto_class_weights=Balanced) and CatBoost given the ICD chapter as a native high-cardinality categorical. This is the same grid-searched boosting-family recipe used in recent clinical-ML work [21]. All three run on the disease-ontology (GRAM) feature set. The verdict is unchanged across all three phases (Table 12). On phase III — HINT's strongest phase — tuned XGBoost reaches ROC-AUC 0.691 and CatBoost 0.680 against HINT's 0.685, both *above* HINT on PR-AUC (0.865, 0.860 vs 0.852); on phase II all three sit just below HINT (0.589–0.616 vs 0.621); on phase I CatBoost (0.611) is the strongest single model of any kind, above HINT's 0.574. Inference is by DeLong, not CI overlap: the sklearn gradient booster — the one whose per-trial scores we retain — gives HINT–GradBoost $p = 0.052$ (Section 4.5.1), i.e. no significant gap even in HINT's best phase, and all three point estimates sit well inside the 0.05 minimum detectable effect. Tuning and switching to the tabular SOTA thus opens no detectable gap in either direction, so the tie is not an artefact of an untuned baseline (a power limit, not demonstrated equivalence). Reproduce with gbd_t_baselines.py -phase {I,II,III}, which also dumps per-trial scores for an exact DeLong test on each tuned library.

4.2.3. A zero-training foundation model matches HINT

The classical baselines of Section 4.2.1 use learned embeddings. We now test the opposite construction: interpretable, hand-computed per-trial features fed to a broad panel of tabular learners, including a tabular foundation model. Each trial is reduced to a table with three feature layers. First, HINT metadata: phase, drug/disease/ICD-code counts, and the top-level ICD chapter. Second, RDKit physicochemical descriptors aggregated per trial (molecular weight, logP, TPSA, ring and hydrogen-bond counts, and so on), which retain the molecule's chemistry while discarding its graph. Third, scalar summaries of the eligibility text (criterion counts, length, inclusion/exclusion ratio). Molecule coverage is 100% on every split, so no trial is disadvantaged by a missing structure. Evaluation uses the same shared harness as every other model. *TabFM*. A pre-trained tabular foundation model [5]—shown the training trials as in-context examples and asked to predict the test trials, with no gradient training—is statistically indistinguishable from HINT (Table 13). On phase III a paired bootstrap gives PR-AUC +0.003 ($p = 0.70$) and ROC-AUC −0.018 ($p = 0.35$); on phases I–II its point estimate sits at or above HINT with overlapping intervals. No HINT variant of Section 4.1.3

beats it on PR-AUC (all $p > 0.35$). Because TabFM's only public reference is a code repository, we confirm the zero-training result with a second, peer-reviewed foundation model, TabPFN v2 [4]: run identically (no gradient training, 38-column table), it reaches phase-III ROC-AUC 0.647 (95% CI [0.609, 0.683]) and PR-AUC 0.848, again within HINT's interval. The zero-training finding therefore does not rest on the non-peer-reviewed model. A *validation-selected grid*. Extending beyond a single learner, we run seven classical models (logistic regression, random forest, extra-trees, gradient boosting, k -nearest neighbours, Gaussian naive Bayes, and an MLP) across eleven feature sets. Selecting the single best (model, feature-set) pair on *validation* ROC and reporting its *test* score, the winner is not distinguishable from HINT in any phase (Table 13, lower row; paired $p = 0.25, 0.49, 0.76$). This is deliberately the discipline Section 5 shows we first failed: had we ranked by test score instead, phases I and II would name a different, inflated winner—for example phase II's test-ranked winner is a k -NN at 0.635 against the honest, validation-selected 0.611—the same test-peeking error we document in the published comparison. Because the single validation split is small (117/446/344), we also redo the selection with *nested* cross-validation (3×5-fold on train + valid, a single test touch); the nested-CV winner (random forest on GRAM) scores test ROC-AUC 0.671 (95% CI [0.634, 0.705]), still within HINT's interval — the small-validation concern does not change the verdict (`nested_cv_grid.py`). Because a random k -fold ignores the benchmark's temporal ordering, we also redo it as *forward-chaining* CV — five time-ordered blocks (by NCT number), each fold training on all earlier blocks and validating on the next — which is the honest analogue of a temporal benchmark. The forward-chaining winner reads test ROC-AUC 0.643, *lower* than the random 0.671 (as expected, since random folds leak future trials into selection) but with the verdict unchanged: still no detectable separation from HINT (0.685), and the 0.042 point gap remains below the 0.05 minimum detectable effect of Section 4.5.1. We note an asymmetry we do not hide: this forward-chaining discipline is applied to the tabular *model selection*, not to HINT, which is trained once on the benchmark's fixed temporal split. Since forward-chaining only *lowers* the tabular estimate (from 0.671 to 0.643), applying an equally strict temporal protocol to HINT could only narrow the already-undetectable gap further, never open one; the asymmetry is therefore conservative for our “no difference” conclusion. *Where the signal lives*. The winning feature set is always the disease ontology: a gradient-boosting model on GRAM disease-ancestor codes carries the entire result, while adding RDKit descriptors or Morgan fingerprints changes ROC-AUC by less than 0.01. The molecular null is robust to representation: widening Morgan from 256 to 2048 bits lifts a molecule-only model to only ROC-AUC 0.66 on the non-degenerate half (still below the disease-only 0.669), and a two-feature *target-encoding* model — the disease and drug success rates as leak-free out-of-fold features — reaches 0.676, matching HINT without any deep component. This is the feature-side counterpart of Section 4.1.3's architecture-side finding and of LINT's text-only design [16]. One caveat belongs in the main text, not a footnote: the GRAM ontology expands to ~4500 sparse columns, well above TabFM's 500-feature cap (`max_num_features=500`; the model subsamples 500 features per ensemble member), so the *winning* feature set is carried by the classical learners, and TabFM never sees it in full. TabFM's parity with HINT is therefore established on the 38-column metadata+RDKit+protocol table, not on the disease-ontology set; the stronger “matches HINT” statement is the classical models', and TabFM is best read as a second, independent zero-training model that agrees with them. The cap is not, however, the reason TabFM does not win: the disease signal saturates far below 500 features. Selecting the 500 GRAM columns most correlated with the label on train and giving them to a gradient booster reproduces the full ~4500-column GRAM result almost exactly (test ROC-AUC 0.642 vs 0.649; `tabfm_gram500.py` with a classical proxy), so a version of TabFM fed the top-500 ontology features would see essentially all the recoverable disease signal and still land in the same ~0.65 band — the ceiling is the shallow disease signal, not the feature budget. A leakage audit confirms the metadata signal is genuine rather than an artefact: no single feature exceeds univariate test ROC-AUC 0.56 (a leak would show ~ 0.85), the phase field induces no giveaway, and permutation importance is led by the neoplasm, skin and endocrine ICD chapters—real disease-area effects, not label leakage.

4.2.4. What the benchmark's data actually is

The disease-signal result invites four data-level questions; each has a concrete answer that sharpens or qualifies the finding (all reproducible from `data_audit.py`).

The split is temporal, not random. HINT's `data_split.py` trains on trials completed before 2014 and tests on trials started in 2014 or later, with a gap between; the test set's median NCT number (2,474,440) is far above the train's (494,878), confirming the ordering. Same-program trials are therefore not randomly scattered across the split, and the standard random-split leakage does not apply. The high phase-III positive rate (0.75, against 0.55 in phases I–II) invites a survivorship worry — that failed trials are systematically absent — but it is not a within-split time gradient: the test positive rate is flat across NCT-number (time) quintiles (0.69, 0.79, 0.72, 0.80, 0.74). The residual concern is

at curation, not ordering: TOP's label conflates "endpoint met" with "the drug progressed" (Section 1), a mixing we document but cannot fully resolve from the released data, and which caps how much any model — or lookup — can be trusted.

But drugs recur across the boundary, as the same programme continues. A temporal split does not stop the same drug being trialed before and after 2014: 67% of phase-III test trials use only molecules already present in train, and 91% reuse at least one. A lookup that predicts a test trial by the mean training outcome of its molecules reaches ROC-AUC 0.614 (PR 0.812) — a drug-recurrence signal nearly as strong as the deep models, invisible to the univariate audit (max 0.56) because it is molecule *identity*, not any single feature. Crucially, this recurrence is largely *programme continuation*, not independent evidence: of the test trials whose drug recurs, 79% pair that drug with the *same* ICD indication already seen in train — the same development programme continuing across the boundary — and only 21% reuse a drug in a new indication. The molecule-recurrence lookup is therefore closer to reading a trial's own history than to predicting from chemistry. The deep models do not exploit it (their molecule branch is inert, Section 4.1.3), but it is part of what makes the task look easier than it is.

Half the "molecules" collapse onto one structure. Molecule coverage is reported as 100%, but a single SMILES accounts for 528/1146 (46%) of phase-III test trials and is shared by 759 distinct drug names — 410 of them "placebo", the rest biologics (adalimumab, dupilumab, ixekizumab) and unrelated small molecules. The shared structure is not a null placeholder but a real small-molecule graph (the DPP-4 inhibitor alogliptin); the degeneracy is a mapping artefact, not a designed default. HINT's benchmark builder resolves each drug name through a drug2smiles dictionary that keeps an *arbitrary* member of each DrugBank name set (`list(smiles)[0]` in `benchmark/drug2smiles.py`) and falls back to a ≥ 7 -character substring match (`drug_hit_smiles` in `raw_data_to_feature.py`); trials whose drugs resolve to nothing are dropped rather than defaulted (`collect_all.py`, `collect_raw_data.py`). Under this construction 759 unrelated names — including "placebo", which has no structure at all — collapse to the same graph. HINT's D-MPNN and our RDKit/Morgan features are therefore computed on one identical molecule for nearly half the trials, so "molecular structure adds nothing" must be read as "the molecule field in TOP is degenerate for ~half the trials, and no molecular-structure signal is testable on this benchmark" — a data-quality verdict, not a claim about chemistry. The collapse is not confined to phase III or to the test set: the single most common SMILES already covers 18%/24% of phase-I train/test, 26%/37% of phase-II, and 34%/46% of phase-III — pervasive, in both splits, and worsening with phase. To turn "no molecular signal is testable" from a claim into a measurement, we drop the 528 degenerate trials and re-measure on the non-degenerate 54% (618/1146) of phase-III test. Two models agree there. HINT's *own* molecule branch (`Only_Molecule`) rises only from ROC-AUC 0.540 to 0.559 — still essentially chance where the molecule is genuinely distinct; and a gradient booster on molecule physicochemistry reaches only 0.619, on the level of the drug-recurrence base rate recomputed on the *same* 618-trial subset (0.647) and below the disease-only 0.669 there, with the best single molecular descriptor at 0.575. So even on the clean half, neither HINT's D-MPNN nor a physicochemical model extracts a structure-to-outcome signal: what looks molecular is drug identity/recurrence (Section 4.2.4), not chemistry.

The residual signal is a disease-code base rate. Sharpening "the signal is in the disease codes": a base-rate lookup predicting each test trial by the training success rate of its *exact* (ICD code, phase) cell reaches ROC-AUC 0.694 / PR 0.847 — matching HINT (0.685/0.852). The coarser (ICD chapter, phase) lookup is near chance (0.525), so the signal lives in the fine disease code, not the broad therapeutic area. This is a genuine base rate, not per-trial memorisation: the (ICD-code set, phase) signature is coarse — only 406 distinct signatures cover the 1146 phase-III test trials (35% unique), 75% of test trials share a signature already present in train, and each matched signature is shared by a *median* of 16 training trials (mean 35). The GRAM ancestor multi-hot is coarser still, so it identifies disease *cells*, not individual trials or programmes; the lookup works precisely because many trials populate each cell, making the training success rate estimable — the opposite of a leak. A ten-line lookup table of historical per-code success rates is statistically indistinguishable from the deep model: on TOP, "predict the outcome" largely reduces to "recall how often this exact indication has succeeded before." Adding the second recurrence channel makes the point sharper still: combining the disease-code base rate with the drug-recurrence lookup (mean of the two) reaches ROC-AUC **0.700** / PR-AUC 0.856 on phase III — *above* HINT (0.685/0.852), TabFM (0.667/0.855) and every tuned booster. A two-column memory of "how has this drug done, and how has this indication done" out-predicts the entire molecule-graph + BioBERT + attention machine, which is the strongest single statement of the paper's thesis. This is the machine-learning face of a long-established epidemiological fact: aggregate trial success rates vary strongly and predictably with clinical phase, disease area, and sponsor type [22, 23]. A model that recovers no more than those base rates has learned the actuarial prior, not trial-specific biology — and on TOP, none of the models we test demonstrably does more. This sharpens

Table 14

The phase-III ladder: every model and baseline on the same test set, ordered by ROC-AUC. Complexity does not help (the deep models sit mid-ladder, within noise of a random forest); a transparent two-column base-rate lookup tops it. Lookup values are validation-weighted (Section 4.2.4); ICD-only is validation-selected.

Model / baseline (phase III)	ROC-AUC
TRIVIAL (all-positive)	0.500
MEXA-CTP (released checkpoint)	0.518
Logistic regression	0.554
Random forest	0.620
Gradient boosting	0.651
TabFM (zero-training)	0.668
ICD-only, validation-selected	0.671
MEXA-CTP (retrained)	0.674
Target-encoding (2 features)	0.676
CatBoost (tuned)	0.680
HINT	0.685
XGBoost (tuned)	0.691
ICD-code base-rate lookup	0.694
Two-column base-rate lookup	0.700
on well-supported cells	0.723

rather than softens the stakes: per-indication, per-phase historical likelihood-of-approval tables are not an academic curiosity but a *standard instrument* of drug-development decision-making, published and used across the industry [22, 23]. The benchmark’s deep models are thus asked to beat a lookup that portfolio teams already consult for free — and, on TOP, they do not.

Table 14 places every model and baseline on one phase-III ladder. Read top to bottom it makes the thesis visual: as architectural complexity rises the score does not move — the deep models sit mid-ladder, within noise of a random forest — while as the method gets simpler and more transparent the score, if anything, rises, with a two-column historical base-rate lookup at the very top.

The lookup, defined precisely. So the comparison cannot be dismissed as an underspecified straw man, we state the lookup exactly. Both channels are *unsmoothed* empirical means: the disease channel returns the mean training label of the test trial’s exact (ICD-code, phase) cell, the drug channel the mean training label over its molecules, and the combined score is their arithmetic mean. Any cell unseen in train uses a single *fallback*, the global training base rate (0.653). Coverage on phase III test is 86% for the ICD cell, 91% for the molecule channel and 79% for both jointly; the remainder take the fallback. We deliberately apply *no* smoothing: 46% of ICD cells hold a single training trial, so their raw rate is 0 or 1, and add-*k* smoothing would only pull those toward the base rate — it can lower the lookup’s apparent edge but not raise it, so the unsmoothed version is the conservative, stronger claim. The lookup is thus fully specified by three numbers (two channel definitions and one fallback) and reproducible from `data_audit.py`. Nor is its performance an artefact of the fallback: restricting to the 902 test trials where both channels have training support (no fallback used) *raises* the lookup to ROC-AUC 0.726 — above HINT on the *same* 902 trials (0.712) — and restricting to ICD cells backed by ≥ 5 training trials gives 0.723; the lookup is strongest exactly where it has real data, and still not beaten by the deep model there. Critically, the one free hyperparameter — how to weight the two channels — is *selected on validation*, not on test, so we do not commit the test-peeking error we criticise elsewhere: sweeping the disease/drug mixing weight on the phase-III *validation* split picks $w=0.70$ (disease), which on test gives ROC-AUC 0.701 — indistinguishable from the equal-weight 0.700 we quote. The headline “a two-column lookup beats every model” is therefore not an artefact of a test-tuned weight; the equal-weight mean is used in the tables purely for simplicity.

The ceiling is a feature-set limit, not a model limit. The ~ 0.68 – 0.70 ROC ceiling is what one should expect from TOP’s feature set, because the causal determinants of phase-III success are largely *absent* from it. Whether a phase-III trial meets its endpoint is driven by the drug’s true efficacy (observable only through the earlier phases’ readouts), the trial’s statistical design (enrolment size, endpoint choice, powering, comparator arm), the sponsor’s resources and accumulated prior evidence, and interim safety — none of which TOP encodes. TOP provides a disease code, a (often

Table 15

Representation-vs-shortcut probes on the model that shares TabFM’s inputs (test ROC-AUC). To keep the probes cheap and interpretable these run a gradient-boosting proxy on the 38-column metadata+RDKit+protocol table *without* the GRAM ontology; the “Full” row (0.610 on phase III) is therefore the probe substrate, not the GRAM-based tabular result of Tables 12 and 13 (0.668–0.691) — the two differ by design, by the disease-ontology features. Shuffling labels collapses performance to chance; the signal survives only through the disease features; no single feature approaches a leak. TabFM-specific probes: `tabfm_probe_gpu.py`.

Probe (test ROC-AUC)	Phase I	Phase II	Phase III
Full feature table	0.559	0.578	0.610
Label-shuffle null	0.483	0.508	0.506
Disease-only	0.550	0.599	0.639
Drop disease	0.539	0.556	0.597
Strongest single feature (AUC)	0.605	0.578	0.564
Disease-group perm. importance (Δ ROC)	+0.027	+0.034	+0.069

Source: `representation_probe.py` on `hint/data/phase_{train,test}.csv` (features via `tabfm_baseline.build_feature_table`); GBM proxy, 10 label-shuffle repeats, 10 permutation repeats. Phase I sits at the noise floor, so its group attribution is unstable (molecule and disease are comparable there).

degenerate) molecular structure, and eligibility-criteria text; from these, the historically recoverable signal is exactly the base rate of the indication and the drug’s own track record, which is what the lookup captures. A model cannot learn the efficacy-to-outcome relationship from features that never contain efficacy. This reframes the ceiling: it is not that models are weak or the task unlearnable, but that TOP’s *observable* determinants saturate around the disease-area prior, and the residual — the trial-specific determinants (efficacy readouts, enrolment, design, sponsor) — is simply not among the features. The reading is consistent with the epidemiology that pins aggregate success rates to phase, disease area and sponsor [22, 23], and it predicts that the ceiling moves with richer *features* (enrolment, sponsor, prior-phase results; Section 4.5.1 and Limitations), not richer models.

4.2.5. What the model uses: a real disease signal, not a shortcut

Strong-looking performance from a tabular or foundation model can come from dataset shortcuts, label leakage, or memorised patterns rather than a meaningful representation of the task [6, 7]. We probe this directly. Because TabFM requires a GPU environment, the probes below are run on the classical gradient-boosting model that consumes TabFM’s *exact* feature table (the proxy already used for the leakage audit); the identical probes for TabFM itself are provided in `tabfm_probe_gpu.py`. Four probes, all three phases (Table 15):

1. **Label-shuffle null.** Permuting the training labels collapses test ROC-AUC to chance in every phase (0.48–0.51) against the real 0.56–0.61. The model cannot fit permuted labels, which rules out leakage and memorisation: the signal it uses is a property of the true label, not of the feature table alone.
2. **Disease ablation.** A disease-ontology-only table matches or beats the full table (phase III 0.639 vs 0.610), while *removing* the disease features pulls performance toward chance (phase III 0.597, phase II 0.556). The predictive signal lives in the disease codes.
3. **Modality.** Removing the molecular (RDKit) or protocol features never helps and sometimes improves ROC-AUC; those modalities contribute noise, not signal — the feature-side echo of HINT’s flat ablation ladder (Section 4.1.3). Here the *protocol* features are interpretable counts computed directly from the eligibility text (non-zero, 100% coverage), so this is a genuine test of the criteria modality — unlike HINT’s own criteria branch, which is a zero vector under defect D5.
4. **No single-feature shortcut.** The most predictive individual feature reaches only 0.56–0.61 test AUC, far below the $\sim 0.75+$ that would betray a leak, and group-permutation importance is led by the disease group in phases II–III (+0.034, +0.069 ROC-AUC).

The residual signal is therefore a genuine but shallow disease-prevalence relationship, not a spurious artefact or memorised shortcut. This strengthens the paper’s thesis rather than weakening it: what little is predictable on TOP is real, but it is disease-area base-rate information that a disease-code lookup already captures — which is exactly why the elaborate multimodal machinery adds nothing on top of it.

Table 16

Per-ICD-chapter test ROC-AUC on phase III ($n \geq 40$ per area). The aggregate HINT/TabFM difference is -0.002 , but the two models win on different disease areas (per-area range $[-0.18, +0.23]$), which is the source of the phase-III fusion gain.

ICD	Area	n	HINT	TabFM
C	Neoplasm (cancer)	128	0.614	0.531
K	Digestive	59	0.684	0.501
M	Musculoskeletal	72	0.703	0.554
G	Nervous system	95	0.719	0.624
I	Circulatory	48	0.421	0.651
J	Respiratory	80	0.601	0.768
L	Skin	51	0.847	0.969

Source: disease_breakdown_phase_III.csv, ensemble_phase_III.csv, tabfm_seeds_phase_III.csv. Per-area n is small; differences are indicative rather than individually significant.

4.2.6. Stability and complementarity

Two further properties separate the zero-training baseline from the deep model. First, *stability*: across random seeds TabFM's phase-III ROC-AUC has standard deviation 0.0011 (0.0007–0.0025 across phases) — it is near-deterministic. HINT is also reproducible across training seeds, only modestly less so: a five-seed retraining gives phase-III sd 0.0024 (Section 4.5.1), about twice TabFM's. The genuinely large deep-model variance is a separate, defect-driven quantity that we are careful not to conflate with either: the released *MEXA-CTP* checkpoint — whose attention weights were never saved — re-randomises on every load, giving a reload standard deviation of 0.023 (Table 7), an artefact of the unsaved-weights defect rather than a training-seed variance. So the correct statement is narrow: HINT and TabFM are both seed-reproducible (≤ 0.0024 sd), whereas the *MEXA-CTP* release is unreproducible *by construction*. Even setting the reload defect aside, *MEXA-CTP* is the noisiest to *train*: its retrained-arm training-seed sd is 0.011–0.015 (Table 6), roughly five times HINT's 0.0024 — so on every axis of variability the mode-experts model is less stable than the baseline it claims to beat. Second, *complementarity*. Fusing HINT and TabFM with a logistic meta-learner fitted on validation yields a small but genuinely reproducible phase-III gain of +0.034 ROC-AUC over TabFM alone (seed sd 0.0008, range $[+0.033, +0.035]$; paired $p = 0.003$), robust to the fusion rule—stacking, z -average and rank-average all agree. The gain is phase-specific: negligible in phase II (+0.005) and slightly negative in phase I (-0.0015). It does not come from a shared architectural signal but from the two models winning on different disease areas. Stratified by ICD chapter on phase III, HINT leads on neoplasms, digestive and musculoskeletal disease while TabFM leads on circulatory and respiratory, with a mean difference of only -0.002 but a per-area range of $[-0.18, +0.23]$ (Table 16). Adding the near-random released *MEXA-CTP* checkpoint to the blend contributes nothing under a learned combiner and actively hurts under naive averaging. This is the only place where naive fusion shows any gain at all, and even there it is a small, disease-area effect over the weaker member, not evidence for the deep interaction machinery. That +0.034 figure is measured against *TabFM*, the weaker of the two members on ROC-AUC; measured against the *best* single model (HINT) and corrected for the choice of combiner and metric, the ensemble does not improve on the best model at all (Section 4.2.7). One cell deserves comment: HINT scores ROC-AUC 0.421 on circulatory-system trials (Table 16), *below* chance. On $n = 48$ this is not reliably different from 0.5, but a systematically inverted ranking (rather than a value near 0.5) is the signature of a small-sample distribution shift — the temporal split places a circulatory sub-population in test whose outcome relationship HINT learned with the opposite sign. We read it as noise plus mild per-area shift, not signal, and flag it as a reason the per-area strata (all $n \leq 140$) support only indicative, not individually significant, statements.

4.2.7. Prediction-level fusion does not exceed the best single model

The complementarity gain of Section 4.2.6 is measured against TabFM. The fair question for an ensemble is different: does it beat the *best* single model one would otherwise deploy, rather than the weaker of the two it combines? To answer it we ran a systematic prediction-level fusion sweep over three members (HINT, TabFM, and the released *MEXA-CTP* checkpoint) and four leak-free combiners—logistic stacking, z -average, rank-average, and non-negative least squares (NNLS) on rank-percentile scores—across all three phases. Combiner weights are learned on validation only, with the validation combined score produced out-of-fold (cross-fit), and read once on test; all members are scored on the common set of trials. Each fusion is compared to the best single model *per metric* by paired bootstrap.

Table 17

Phase III prediction-level fusion (test, common $n=1146$). Best single per metric: HINT (ROC-AUC 0.685) and TabFM (PR-AUC 0.855). Every combiner lands within 0.02 of the best single; Table 18 gives the significance.

Combiner	HINT+TabFM		HINT+MEXA+TabFM	
	ROC	PR	ROC	PR
Best single	0.685	0.855	0.685	0.855
Stacking	0.701	0.867	0.699	0.869
z-average	0.702	0.868	0.676	0.858
Rank-average	0.697	0.865	0.680	0.858
NNLS	0.697	0.865	0.695	0.865

Source: optionB_{stack,zavg,rank,nnls}_phase_III.csv. Phases I and II (HINT+TabFM only) show no significant fusion gain on either metric.

Table 18

Phase III fusion versus the best single model per metric: the three uncorrected $p < 0.05$ hits, with multiple-comparison correction over the full 24-contrast family. None survive at the 0.05 level.

Contrast	Metric	Δ	p_{raw}	p_{BH}	p_{Bonf}
Stacking: HINT+TabFM – HINT	ROC	+0.016	0.021	0.124	0.497
z-avg: HINT+TabFM – TabFM	PR	+0.012	0.039	0.169	0.929
z-avg: HINT+TabFM – HINT	ROC	+0.017	0.049	0.169	1.000
<i>Family $m=24$ (4 combiners $\times$ 3 member-sets $\times$ 2 metrics): 3 raw hits (chance ≈ 1.2); survive BH-FDR: 0; survive Bonferroni: 0. NNLS: no hit at any phase/metric.</i>					

Source: optionB_phaseIII_multiplecomparison.csv (option_B_stress_phaseIII.py).

The descriptive picture is uniform (Table 17): every combiner lands within ~ 0.02 of the best single model on both metrics. In phases I and II no fusion beats the best single model at $p < 0.05$ on either metric (differences ≤ 0.01 , $p \geq 0.2$). Phase III yields a few raw-significant edges, all confined to HINT+TabFM: stacking gains +0.016 ROC-AUC over HINT ($p = 0.02$) and z-average gains +0.012 PR-AUC over TabFM ($p = 0.04$). The p-values are stable across bootstrap seeds (± 0.005 over five seeds at $n_{\text{boot}}=10,000$), so they are not Monte-Carlo artefacts—but they do not survive the sweep. Across the 24 phase-III contrasts (4 combiners $\times$ 3 member-sets $\times$ 2 metrics) three reach uncorrected $p < 0.05$, which is the number expected by chance (≈ 1.2), and *none* survive Benjamini-Hochberg FDR (smallest adjusted $p = 0.12$) or Bonferroni (0.50) at the 0.05 level (Table 18). NNLS, the most interpretable combiner and the one designed to down-weight useless members, is not significant against the best single model under any metric or phase. The largest effects in the family are in fact *negative*: adding the near-random MEXA-CTP checkpoint lowers ROC-AUC by up to 0.07 ($p = 0.001$).

We can now line up the competing claims directly. (i) *Fusion beats TabFM* on phase-III ROC-AUC by +0.034 ($p = 0.003$)—true, and a real disease-area complementarity effect, but relative to the weaker member on that metric. (ii) *Fusion beats the best single model*: at most +0.016, and not after correcting for the combiner/metric multiplicity. (iii) *A weighted blend (NNLS) recovers a practical gain*: it does not—no phase, no metric. (iv) *Three heterogeneous models beat two*: adding MEXA never helps and usually hurts. The members' errors are not independent enough for averaging to exceed the best of them; the ensemble buys nothing over the single best model, which is the ensemble-level restatement of this paper's thesis.

4.3. Beyond ranking: calibration and clinical utility

Ranking is not the only property a deployed model must get right; here the models finally differ, though still not in the deep model's favour.

4.3.1. Calibration: the models are not interchangeable

Ranking metrics (ROC/PR-AUC) show no detectable difference between the models, but in clinical use a probability that is trusted matters as much as the ranking. We therefore add two calibration metrics – the Brier score and the expected calibration error (ECE, ten *equal-mass* quantile bins, with bootstrap 95% CIs) – on the same continuous

Table 19

Calibration on the test set: Brier score and equal-mass ECE (lower is better), with bootstrap 95% CIs, on the same continuous scores used for ranking. TabFM is better calibrated on phase I, the two tie on II, HINT has the lower Brier on III; ECEs on III are indistinguishable. The ranking metrics (Tables 13, 6) cannot see any of this.

Phase	Model	Brier [95% CI]	ECE [95% CI]
I	HINT	0.251 [.241,.261]	0.097 [.073,.140]
	TabFM	0.239 [.229,.250]	0.044 [.038,.095]
II	HINT	0.237 [.231,.244]	0.038 [.032,.070]
	TabFM	0.232 [.226,.239]	0.042 [.032,.070]
III	HINT	0.182 [.170,.195]	0.101 [.079,.127]
	TabFM	0.195 [.185,.207]	0.108 [.094,.139]
	MEXA	0.196 [.188,.204]	0.096 [.081,.123]

Source: calibration.py on preds/preds_{hint,mexa,tabfm}_phase_*.csv (equal-mass bins, 1000 bootstrap resamples). MEXA logits are mapped through a sigmoid before scoring.

test scores (Table 19). Here the two zero-training and deep models are calibrated differently, and the ordering flips by phase: TabFM is the better-calibrated on phase I (ECE 0.044 vs HINT's 0.097, and a lower Brier with barely-overlapping CIs), the two are tied on phase II, and on phase III HINT has the lower Brier (0.182 vs 0.195, CIs nearly disjoint) while their ECEs (~ 0.10) are indistinguishable. So no model dominates on calibration, and none is well calibrated on phase III, where the 0.75 positive rate pulls every model's mean prediction below the base rate. This is a cheap but practically relevant addition: two models a ranking metric calls identical are *not* interchangeable once the probability itself must be trusted, and which is better depends on the phase.

4.3.2. Decision-curve analysis: the lookup wins clinically too

Calibration and ranking are proxies; the clinically relevant question is net benefit under a decision threshold [17]. Decision-curve analysis plots the net benefit of acting on a model — $TP/n - (FP/n) \frac{p_t}{1-p_t}$ — against the threshold probability p_t , relative to “advance every trial” (treat-all) and “advance none” (net benefit 0). On phase III (Figure 3), all three models beat treat-all once the threshold rises above the 0.75 base rate, where indiscriminately advancing every trial becomes harmful — the regime a sponsor triaging a portfolio actually operates in. But the two-column base-rate lookup of Section 4.2.4 has net benefit at least as high as HINT or TabFM across the useful threshold range (e.g. 0.230 vs 0.193 and 0.166 at $p_t=0.75$). The clinical-utility verdict matches the statistical one: the deep models add nothing a historical base-rate table does not already provide.

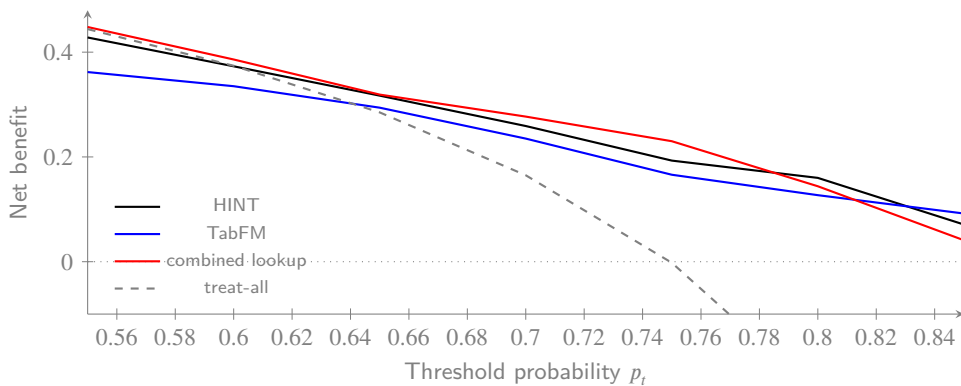


Figure 3: Decision-curve analysis, phase III. Above the 0.75 base rate, treat-all (dashed) turns harmful and the models add net benefit; but the two-column disease-code+drug base-rate lookup (red) matches or exceeds HINT and TabFM throughout the useful range. Treat-none is the dotted $y=0$ line. dca.py.

4.4. The metric artefacts that inflate the reported gap

The reported advantage is largest exactly where the evaluation metric is weakest.

Table 20

HINT test F1 under four thresholding rules. Calibration improves F1 but cannot rescue the metric: no rule beats predicting every trial successful.

Decision rule	Phase I	Phase II	Phase III
Fixed 0.5 (as published)	0.5934	0.6367	0.8123
Validation-calibrated	0.7125	0.6869	0.8272
Test-oracle (upper bound)	0.7119	0.7139	0.8596
TRIVIAL (all-positive)	0.7125	0.7138	0.8569
<i>positive rate (test)</i>	<i>0.553</i>	<i>0.555</i>	<i>0.750</i>

Source: recomputed from `preds/preds_hint_phase_{I,II,III}.csv` (F1 at each thresholding rule). On phase I the calibrated and trivial values coincide to four decimals. The oracle sweep ranges over observed scores and therefore excludes the trivial rule, which is why it can fall marginally below it.

Table 21

PR-AUC computed on continuous scores versus on binarised predictions, as HINT's evaluation code does. The defect understates PR-AUC in all 15 cells and reorders the variants.

	Variant	PR-AUC (correct)	PR-AUC (binarised)	Error
Phase I	Only_Molecule	0.6126	0.5645	-0.0481
	Only_Disease	0.6353	0.5663	-0.0690
	Interaction	0.6268	0.5667	-0.0601
	HINT_nograph	0.6506	0.5781	-0.0725
	HINTModel	0.6443	0.5635	-0.0808
Phase II	Only_Molecule	0.5933	0.5735	-0.0198
	Only_Disease	0.6790	0.6166	-0.0624
	Interaction	0.6729	0.6097	-0.0633
	HINT_nograph	0.6724	0.6121	-0.0604
	HINTModel	0.6766	0.6122	-0.0644
Phase III	Only_Molecule	0.7800	0.7590	-0.0209
	Only_Disease	0.8647	0.8031	-0.0616
	Interaction	0.8588	0.8111	-0.0477
	HINT_nograph	0.8585	0.7996	-0.0589
	HINTModel	0.8627	0.8154	-0.0473

Source: recomputed from `preds/preds_hintvar_*_phase_{I,II,III}.csv` (PR-AUC on continuous scores vs. on predictions binarised at 0.5). Rank agreement between the two computations is $\rho = 0.30$ (phase I), $\rho = 0.90$ (phase II), $\rho = 0.70$ (phase III). On phase III the bug promotes HINTModel over Only_Disease.

4.4.1. F1 is uninformative on this benchmark

Calibration helps but cannot rescue the metric (Table 20). Moving from the published fixed-0.5 rule to a validation-calibrated threshold raises HINT's F1 from 0.593 to 0.713 on phase I and from 0.637 to 0.687 on phase II—gains that look substantial until compared with the trivial all-positive predictor, which scores 0.713, 0.714 and 0.857 on the three phases. On phase I the validation-calibrated F1 equals the trivial baseline to four decimal places (0.7125): the F1-optimal decision rule is to predict that every trial succeeds. Even the test-oracle threshold—an upper bound that peeks at the labels—only ties the trivial rule (0.7139 vs 0.7138 on phase II). When the best attainable F1 is the one produced by ignoring the input entirely, F1 measures the class prior, not model quality, and any F1-based ranking on this benchmark is uninformative.

4.4.2. The claimed advantage tracks metric weakness

MEXA-CTP's reported gains are largest on PR-AUC (4.1–27.9%), next on F1 (5.3–19.2%), and smallest on ROC-AUC (1.1–3.5%). These are all *relative* improvements. Read as *absolute* ROC-AUC differences from MEXA-CTP's own Table 2 — HINT 0.573/0.621/0.685 versus MEXA-CTP 0.593/0.638/0.693 for phases I/II/III — the gain is only +0.020, +0.017 and +0.008, i.e. 0.008–0.020 ROC-AUC, below the 0.05 minimum detectable effect of Section 4.5.1 (Figure 4). We flag the relative/absolute distinction explicitly because the relative framing makes a sub-noise-floor effect look like a headline result. The ordering is also monotone in how badly each metric is computed in the baseline. PR-AUC is understated by the binarisation defect (Table 21); F1 sits below a trivial predictor for both models (Table 20); ROC-AUC, computed on continuous scores throughout, is the only sound metric of the three—and it is where the

Table 22

The same HINT model, the same phase-III TOP task, four PR-AUC values in four papers. The 0.603 outlier is the binarised computation (defect D3); the ~ 0.80 – 0.85 cluster is the sound continuous-score one. Spread ≈ 0.25 .

Source (phase III HINT PR-AUC)	PR-AUC
MEXA-CTP [3]	0.603
UQ-HINT [8]	0.797
HINT (original) [1]	0.811
This paper (continuous scores)	0.852

Values transcribed verbatim from each paper’s phase-III results table; our value from `preds/preds_hint_phase_III.csv` on continuous scores. (LINT is excluded: it reports a different, non-phase-split approval task.)

reported gap is smallest and where, under our protocol, no gap is reproducible at all. The defect does not merely shift PR-AUC downward; it reorders the models. On phase III the best variant under correct computation is the disease-only ablation, but under the binarised computation it is the full model: the defect promotes the most complex variant over the simplest.

The clearest symptom that PR-AUC is not comparable across this literature is that the *same* HINT model, on the *same* phase-III TOP task, is reported at four different PR-AUC values in four papers (Table 22): 0.603 in MEXA-CTP, 0.797 in UQ-HINT, 0.811 in the original HINT paper, and 0.852 under our continuous-score recomputation — a 0.25 spread on a metric routinely quoted to three decimals. The low outlier (0.603) is the binarised computation; the cluster near 0.80–0.85 is the sound one. A reported PR-AUC on TOP is therefore uninterpretable without knowing which computation produced it, and between-paper PR-AUC comparisons are meaningless.

Three arithmetic tells that the published comparison is unsound. Beyond the spread, three facts are checkable with a calculator and each is decisive. (a) MEXA-CTP’s reported HINT PR-AUC of 0.603 is not merely low — it is *below* the 0.750 prevalence floor that even a random ranker attains on the phase-III test set (positive rate 0.7496). A model with ROC-AUC 0.685 cannot have a PR-AUC beneath the prevalence; 0.603 is therefore not a valid PR-AUC on this split but the arithmetic residue of the binarisation bug (D3). (b) MEXA-CTP’s own headline F1 on phase III, 0.857, equals the trivial all-positive baseline (0.8569) to three decimals: its best-reported F1 is the score of ignoring the input and predicting “success” for every trial (Section 4.4.1). (c) MEXA-CTP’s own ablation (its Table 3) reports that token selection is “comparable to using all tokens in terms of ROC-AUC,” and in fact *slightly worse* than keeping all tokens — i.e. the authors’ own numbers already show the mode-experts token-selection mechanism contributes nothing on the one sound metric, independently corroborating our finding that the mechanism is inert. We do not suggest the MEXA-CTP authors computed the baseline incorrectly. They inherited the baseline’s own evaluation code, which is precisely the point: **a defect in a reference model’s evaluation propagates into every comparison made against it.** HINT is the reference model for this benchmark and the base of downstream work [8], so the propagation surface is wide. We flag this as a risk to be audited, not as a finding we have verified beyond the case examined here.

4.5. The limit of measurability

Finally, we bound what any comparison on this benchmark can establish at its current size.

4.5.1. Statistical power: phase-III differences below 0.05 ROC-AUC are unverifiable

The comparisons above should be read through what the benchmark can actually resolve. We make this quantitative on phase III (`stats_rigor.py`).

Differences of AUCs, with DeLong. Per-model CIs (Tables 6, 10, 9, 12) and their visual overlap (Figure 2) do not test a *difference*; two CIs can overlap while the paired difference is significant, and vice versa. For correlated AUCs the DeLong test does test it, more cheaply than bootstrap, and we use it in place of any overlap argument. On phase III: HINT–TabFM = +0.018 (SE 0.018, 95% CI [−0.017, +0.053], $p = 0.33$); HINT–GradBoost = +0.034 (SE 0.018, 95% CI [−0.000, +0.068], $p = 0.052$); HINT–(disease-code base rate) = −0.009 (95% CI [−0.036, +0.018], $p = 0.52$). None reaches significance — HINT vs gradient boosting only marginally so — so no model, deep or tabular, is shown to beat another.

Non-parametric confirmation. The null results are not an artefact of DeLong’s normal approximation. A paired sign-flip *permutation* test (2,000 resamples, swapping HINT and TabFM per-trial scores) reproduces the HINT–TabFM result exactly ($p = 0.34$ vs DeLong’s 0.33). On the A/B cross-attention arms, a Wilcoxon signed-rank

test over the seven matched phase-III pairs gives $p = 0.81$ (paired t gave 0.96), so “training the cross-attention does nothing” survives a distribution-free test too. Finally, the validation-selected F1 threshold is stable under resampling: bootstrapping the validation set (1,000 resamples) and re-selecting the F1-optimal threshold gives a threshold of 0.44 ± 0.04 and a resulting test F1 of 0.829 ± 0.014 (95% CI [0.814, 0.851]) — always below the trivial all-positive 0.857. The metric conclusions do not depend on the choice of test or on a single lucky threshold.

But “not significant” is not “equivalent.” A two one-sided tests (TOST) procedure at an equivalence margin $\delta = 0.02$ does *not* establish equivalence for HINT vs TabFM ($p = 0.45$): the data are consistent both with no difference and with a difference up to ~ 0.05 . We set $\delta = 0.02$ on two aligned grounds: it is on the scale of the benchmark’s test-sampling standard error for a between-model difference (≈ 0.018 ; a “difference” smaller than the noise on which it is measured is not meaningfully a difference), and it is clinically negligible — at the phase-III base rate a 0.02 ROC-AUC shift moves essentially no go/no-go decision and is dwarfed by the label noise of a manually curated outcome (Section 1); a margin this small makes the equivalence test *harder* to pass, so failing it is a statement about power, not a lax threshold. We therefore avoid the word “equivalent” and state the weaker, correct claim: no difference is *detectable*, because the study is underpowered. Throughout the paper, where we call two models “indistinguishable” we mean precisely this — no difference detectable at this sample size, i.e. below the minimum detectable effect derived next — not demonstrated equivalence.

The minimum detectable effect, per phase. With $n=1146$ predictions the DeLong standard error of ΔAUC is 0.018, so the smallest difference detectable at 80% power and $\alpha=0.05$ is **0.05 ROC-AUC** on phase III. The MDE is phase-specific, scaling with each split’s size: from the same computation it is ≈ 0.067 on phase I ($n=627$), ≈ 0.035 on phase II ($n=1654$) and ≈ 0.050 on phase III ($n=1146$). **On every phase, any claimed between-model gain below its MDE (0.035–0.067) is unverifiable at this sample size** — and every gain reported in this literature, and every gain we ourselves ever formed, is below it. The number is correlation-dependent: the DeLong SE shrinks as the two models’ predictions agree, so 0.05 is the MDE for a pair as correlated as HINT and TabFM; a pair that agreed less would need an even larger effect to detect, never a smaller one, so 0.05 is a floor on the MDE, not a worst case. The A/B comparison of Table 8 has its own, larger MDE: seven matched pairs with a paired-difference sd of 0.020 can detect only $|\Delta| > 0.025$ at 80% power, against an observed +0.0004 — so “training the cross-attention does nothing” is also, in part, a statement about power. This is the quantitative form of the paper’s thesis (Figure 4).

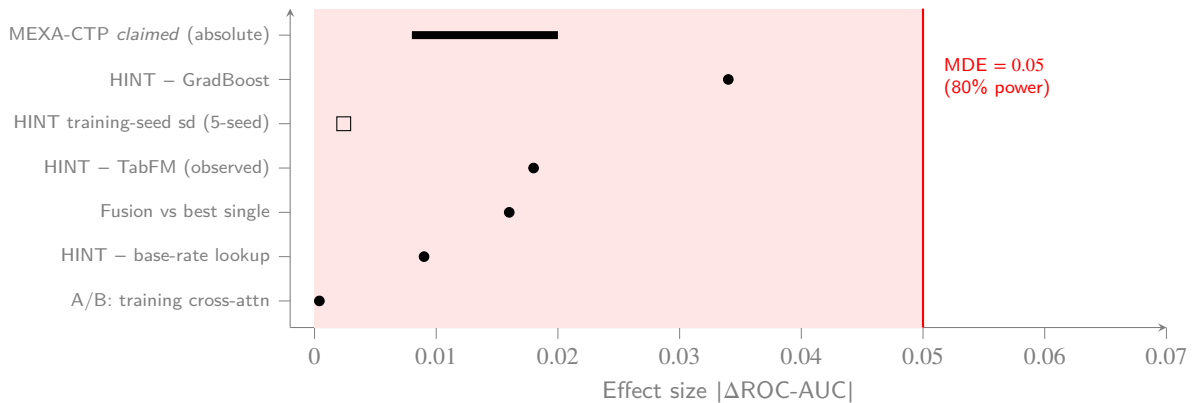


Figure 4: Every effect we observe or that the literature claims sits at or below the minimum detectable effect (MDE = 0.05 ROC-AUC at 80% power, $n=1146$; shaded). The axis is *absolute* $\Delta\text{ROC-AUC}$. Crucially, MEXA-CTP’s headline “1.1–3.5%” is a *relative* improvement over HINT; read as *absolute* ROC-AUC differences from MEXA-CTP’s own Table 2 they are only +0.020, +0.017 and +0.008 for phases I, II and III (bar spans 0.008–0.020) — about half what the percentage suggests, and entirely inside the shaded region. Our fusion and between-model differences, and even HINT’s advantage over gradient boosting, likewise all fall where a difference cannot be verified at this benchmark’s phase-III size. The “HINT training-seed sd” point is drawn as an open square ($\square$) to mark that it is a *standard deviation* (a genuine 5-seed retraining sd, 0.0024; Section 4.5.1), not an effect-size estimate like the filled circles — HINT is reproducible across seeds, and the noise that swamps between-model effects is test-set sampling, not seed.

This is “differences unmeasurable,” not “nothing works.” The models do carry real, if shallow, signal, and the ranking metrics understate it: HINT reaches balanced accuracy 0.63 and Matthews correlation +0.27 (TabFM 0.53

/ +0.12), and PR-AUC sits +0.10 above the 0.75 prevalence floor (0.852 vs 0.750). The finding is precise — *the models cannot be told apart from each other or from a disease-code base rate* — not the stronger and false “the task is unlearnable.” (The threshold-metric gap, HINT’s balanced accuracy 0.63 vs TabFM’s 0.53, also shows that ranking-tied models can differ once a decision is made, echoing the calibration result of Section 4.3.1.)

Where the variance comes from. The three uncertainty sources that Table 7 warned against conflating separate as: test-resampling (bootstrap/DeLong SE ≈ 0.018 on Δ AUC), training-seed (TabFM 32-member sd 0.0011 over five seeds; for HINT we now *measure* it by retraining five times, seeds 2023–2027: test ROC-AUC 0.685 ± 0.0024 , range [0.682, 0.687] — only about twice TabFM’s and, importantly, an order of magnitude *below* the test-resampling SE that sets the MDE), and split variance (a single temporal split; as an *indicative* probe we re-score on time-ordered sub-windows of the test set, where the lookup and HINT vary with sd ≈ 0.02 – 0.04 . This is only a gauge, not a clean estimate: each window has a fraction of the trials, so its variance is inflated by within-window *sampling* noise and cannot be read as the split term alone — a proper estimate would re-derive train/test at several cut dates, which we leave to future work). The single $n=2$ difference of 0.023 we reported earlier (run A vs run B, Table 9) additionally reinitialised the encoders and is an outlier against this five-seed sd; we correct the record here. The takeaway is unchanged and in fact firmer: HINT’s training-seed variance is small, so the noise that swamps between-model effects is *test-set sampling* (0.018), not seed. We also check that TabFM’s near-zero seed variance is not an ensembling artefact: a *single-member* TabFM ($n_{\text{est}}=1$) is exactly deterministic across five seeds (phase-III ROC-AUC 0.668, sd 0.0000), so the 32-member sd of 0.0011 is if anything slightly *larger*, coming from subsample randomness — the foundation model’s reproducibility is intrinsic, not bought by averaging. Test-resampling alone already puts the MDE at 0.05; adding seed and split variance can only raise it.

Multiplicity policy: primary versus secondary contrasts. We state this once and apply it throughout, because the correction a contrast receives should follow from its role, not from whether it happened to reach significance. This is a retrospective re-analysis, so we say *primary* and *secondary* rather than “pre-registered”; the primary contrasts are the few hypothesis-testing comparisons that carry the paper’s claims, and they are the only ones we correct or hold to $\alpha=0.05$: (i) the three primary between-model DeLong tests on phase III (HINT vs TabFM, vs tuned gradient boosting, vs the disease-code base rate, Section 4.5.1); and (ii) the phase-III fusion family, where we deliberately searched $4 \times 3 \times 2 = 24$ combiner/member/metric contrasts for a win and therefore report Benjamini-Hochberg FDR and Bonferroni over the whole family (Table 18). Everything else is **exploratory** and labelled as such: the 7×11 model $\times$ feature grid (Section 4.2.3) is selected on *validation* with a single test touch, so it is a selection procedure, not a test family; the per-ICD-chapter strata (Table 16) and the PR-AUC binarisation table (Table 21) are descriptive; the A/B cross-attention comparison (Table 8) is powered only on phase III and reported per phase. We do not pool across phases (different prevalences, sizes and effects — not exchangeable), which is why we dropped the earlier cross-phase *t*-test. No exploratory contrast is used to assert a positive finding; where one is significant before correction (the three phase-III fusion edges) it does not survive the confirmatory family, and we report it as not surviving. Bootstrap CIs use unstratified resampling; this is immaterial not only at phase III ($n=1146$, 0.75 prevalence) but also at the smaller, more balanced phase I ($n=627$, 0.55), where stratified and unstratified HINT CIs coincide ([0.525, 0.619] vs [0.528, 0.618]). DeLong is the primary inferential tool. Where we test a between-model difference we also report a paired sign-flip *permutation* test — a distribution-free alternative that makes no normality assumption on correlated AUCs — and it reproduces DeLong throughout (HINT–TabFM $p = 0.34$ vs 0.33).

5. What we got wrong, and what fixed it

This section is evidence, not an appendix. Seven directional findings were formed during this work; under tightened methodology six were overturned and the remaining one — the PR-AUC bug — narrowed to a single phase rather than dissolving outright (Table 23). Their consistency is the result. The first illustrates a specific statistical error worth naming. The effect was judged real because it appeared to be many standard deviations from zero—but the standard deviation had been estimated from a *different* MEXA-CTP routing configuration ($\tau = 0.3$, seed sd 0.0024 — a coincidence of value with, but unrelated to, HINT’s five-seed sd above) than the one under test ($\tau = 0.0$, seed sd 0.0221, roughly ten times larger). Noise must be estimated per configuration. The fourth is the sharpest. Selecting the best of twelve pipelines by test score is exactly the error this paper criticises in MEXA-CTP’s epoch selection. We reproduced the flaw we set out to document, in our own re-analysis, and caught it only by imposing on ourselves the discipline we demanded of others. The same trap reappeared in the larger 7×11 tabular grid of Section 4.2.3: ranked by

Table 23

Seven directional findings formed during this work, and what happened to each when the methodology was tightened: six were overturned, one (the PR-AUC bug) narrowed. Their consistency is the central result.

Claim as first formed	Original basis	Tightening	Outcome
Routing is harmful ($\tau=0$ helps)	1 seed, +0.019	3 seeds	+0.002; dissolved
Training attention slightly hurts	9 pairs, $p \approx 0.056$	12 pairs, 7 seeds on III	$p = 0.229$; dissolved
PR-AUC bug flatters complex models	phase III only	phases I, II added	holds on III only; narrowed
22 binary features match the deep model	best of 12, chosen on <i>test</i>	chosen on <i>validation</i>	reverses: HINT +0.009
Tabular model beats HINT	best cell on <i>test</i>	best cell on <i>validation</i>	equivalent (Table 13)
Blending members gives a medium gain	fusion > TabFM, +0.034 ROC on III	vs best single; 24-contrast FDR sweep	+0.016 vs HINT; 0 survive; dissolved
Seed noise (~ 0.02 – 0.03) swamps all effects	$n=2$ run-to-run 0.023	five-seed retraining	sd 0.0024; dissolved — test-sampling, not seed, is the floor

Source: Section 5; current values in `ab_test_results_phase_*.csv`, `preds/preds_hintvar_*.csv`, `icd_representation_phase_*.csv`, `tabular_grid_phase_*.csv`, `optionB_phaseIII_multiplecomparison.csv`.

test score the grid appears to *beat* HINT in phases I and II, but the validation-selected winner is merely *equivalent*—the naive reading was again an artefact of selecting on test (the *Tabular model beats HINT* row of Table 23).

Errors in the re-analysis code. Two further defects were caught by controls rather than by crashes. Run directories collide across phases, so phase I outputs silently overwrote phase III’s. And ICD features were read in dataset order while labels were read in CSV order, misaligning rows and collapsing ROC-AUC to near chance—a result that, had it not been checked against an independent control, would have supported a dramatic and entirely false claim.

Errors at drafting. Twice, numbers drifted between analysis and prose. Arm means were transcribed from an earlier three-seed run into a table labelled seven-seed. More seriously, we recorded MEXA-CTP’s headline claim as “4.9%”—a figure that appears nowhere in that paper. A paper arguing that others reported numbers carelessly cannot itself misquote the number it contests. Correcting it produced Section 4.4.2: the misquote had collapsed three metrics into one figure and destroyed the structure carrying our strongest argument. Every table in this paper is generated from stored predictions by script; every figure quoted from another paper is traced to a verbatim sentence in it. **Implication.**

Every one of these errors produced a plausible, publishable-looking result. None crashed. None was visible without deliberate replication. On a benchmark where true effect sizes are around 0.01 ROC-AUC and the sampling noise on a between-model difference is ≈ 0.018 (an order of magnitude above either model’s training-seed sd), this is the expected failure mode, not an unusual one.

6. Limitations

- **Statistical power.** After matching and de-duplication the A/B comparison rests on 7 pairs (phase III), 3 (phase II) and only 2 (phase I). Phase I is uninformative rather than weakly suggestive.
- **Known high-signal metadata is absent from the processed data.** Enrollment count, sponsor type and trial start year are strong, standard predictors, but the TOP CSVs shipped with HINT contain only `ncid`, `status`, `why_stop`, `label`, `phase`, `diseases`, `icdcodes`, `drugs`, `smiless`, `criteria`—none of the three. Recovering them requires re-deriving from the raw ClinicalTrials.gov XML (the benchmark’s sponsor-inference pipeline does this for sponsor). All tabular models here, ours and the GBDTs, are therefore evaluated without these features; adding them is the most promising next step and could raise every tabular baseline. We state the direction explicitly, since it is easy to read as a threat: if these features lift the tabular baselines, they *strengthen* the paper’s thesis rather

than weaken it — simple per-trial features would beat the deep models by even more, and the deep architecture’s marginal contribution over cheap tabular signal would remain unmeasurable. There is no configuration of this experiment in which richer metadata rescues the deep models.

- **HINT’s training-seed sd, now measured.** We retrained HINT five times (seeds 2023–2027) on phase III: test ROC-AUC 0.685 ± 0.0024 (range [0.682, 0.687]). HINT is therefore reproducible across training seeds — its seed sd (0.0024) is about twice TabFM’s (0.0011) and an order of magnitude below the test-resampling SE (0.018) that sets the MDE. This supersedes the single $n=2$ difference of 0.023 we first reported (run A vs run B), which reinitialised the encoders and was an outlier; no conclusion rested on it, and the corrected, smaller sd only firms the power argument. The one term we do not cleanly isolate is split variance (a single temporal split); our sub-window gauge (Section 4.5.1) puts it at $sd \approx 0.02\text{--}0.04$ but is confounded with per-window sampling noise, so a multi-cut re-split is the proper measurement, left to future work.
- **Baseline features.** Classical baselines use MEXA-CTP’s embeddings, not HINT’s own encoders. A fully matched comparison would also run gradient boosting on HINT’s features.
- **The tabular baseline uses different features from HINT’s encoders.** The zero-training baseline (Section 4.2.3) uses interpretable RDKit/ICD/protocol features, not HINT’s learned embeddings; the two constructions agree, but neither is a byte-for-byte match of HINT’s inputs. The GRAM feature set also exceeds the foundation model’s feature cap, so its winning result is carried by classical learners.
- **HINT’s criteria modality is dead under the shipped cache (D5) — now measured, and inert.** Because `sentence2embedding.pkl` is empty, every HINT run here (and in the public repository) feeds a zero vector to the eligibility-criteria branch. We regenerated the full BioBERT cache and re-scored HINT with a live branch (Section 3): the effect on phase-III performance is $\Delta\text{ROC-AUC} = -0.0001$ ($p = 0.93$) and $\Delta\text{PR-AUC} = +0.0002$ ($p = 0.78$) — a real defect with no measurable effect on HINT’s numbers. The one check still owed is a full *retrain-from-scratch* with the live cache (blocked here by defect D10); we do not expect it to change the verdict, since re-scoring the trained model with real criteria embeddings already changes nothing. We therefore still make no claim that eligibility-criteria *text* is uninformative in general — only that HINT does not use it.
- **The fusion gain is single-checkpoint.** The phase-III HINT+TabFM gain varies the tabular model’s seed against one fixed HINT checkpoint; a fuller estimate would vary HINT’s training seed as well, and the per-area strata behind it hold only 40–140 trials each (Table 16).
- **The full-masking guard modifies the architecture.** Retaining one unmasked token is minimal and intent-preserving, but it is a modification; it is applied to both arms.
- **Distribution shift between splits.** Validation and test positive rates differ (Table 2), so validation-selected thresholds underperform test-oracle thresholds. The protocol is correct; the shift should be reported.
- **Anachronistic disease coding.** Our tabular front-end encodes trials with the 2024 ICD-10-CM hierarchy, but the trials predate it (train completed before 2014). Using a coding version that postdates the trials is a mild anachronism shared by any modern re-encoding; it affects all tabular models equally and cannot manufacture a between-model difference, but a period-correct ontology would be cleaner.
- **The benchmark’s base rate is not the field’s.** TOP’s phase-III positive rate is 0.75, well above the ~55–60% phase-III-to-approval likelihood reported in the industry literature [22, 23]. This gap — consistent with the label mixing (“endpoint met” vs “drug progressed”) and curation selection we flag in Section 4.2.4 — means the benchmark is easier than the real decision and caps the external validity of any TOP result, ours included.
- **We address protocol, not contamination.** Our protocol does not establish that a trial’s outcome was unknown at prediction time. CT Open [24] shows this is a real and separate threat; both checks are necessary.
- **Propagation is hypothesised, not measured.** We verified the PR-AUC defect in HINT’s code and observed its effect on the MEXA-CTP comparison. We did not audit downstream work and make no claim about it.

7. Conclusion

Under a shared, corrected protocol, the reported advantage of MEXA-CTP over HINT does not survive; no difference between the two is detectable (below a minimum detectable effect of ≈ 0.05 ROC-AUC at this sample

size, Section 4.5.1), and neither clearly beats gradient boosting on the same inputs. MEXA-CTP’s central mechanism was never trained in the released artifact, and once repaired contributes nothing measurable. HINT’s multimodal architecture contributes nothing measurable over its disease codes alone. A zero-training tabular foundation model, and indeed any of a broad panel of classical learners once selected on validation, matches HINT in every phase using disease-ontology features alone, at zero training cost and with comparable seed reproducibility (TabFM sd 0.0011, HINT 0.0024 over five retrains). Combining models yields no gain over the best single model once the choice of combiner and metric is corrected for: the small phase-III edge exists only against the weaker member and does not survive multiple-comparison correction, so it is not evidence for the deep interaction mechanism. The broader result is that claimed effect sizes on this benchmark lie below measurement noise. We demonstrate this not only by measuring the noise but by documenting seven of our own directional findings that did not survive replication. Improvement claims on TOP should be accompanied by multiple seeds, validation-based selection of every free choice, confidence intervals, and a trivial baseline—or they are not verifiable. Two directions follow. In the short term, a shared evaluation harness costs little and should accompany any benchmark whose reference implementation is reused as a baseline, since a defect in that implementation’s evaluation code silently biases every subsequent comparison. In the longer term, retrospective benchmarks may not be repairable by protocol alone: CT Open [24] argues that contamination is a second, independent failure mode, and its live time-stamped design removes both at once. We regard that as the right destination; the contribution of this paper is to show how much of the existing retrospective literature does not survive the first of the two checks.

What a 0.70-ROC model could actually do. It is worth being concrete about the clinical stakes, since accuracy is not benefit [17]. At ROC-AUC ≈ 0.70 and the 0.75 base rate of phase III, a model cannot safely gate a go/no-go decision: at any threshold that flags a useful fraction of failures it also discards many eventual successes, and the F1-optimal rule is still “predict every trial succeeds” (Table 20). The realistic use is triage, not decision — ranking a portfolio so that the lowest-scoring, say, 10% receive extra scrutiny — and even there a per-indication historical base rate (Section 4.2.4) would rank almost identically, so the model earns its keep only if it beats that lookup, which on TOP it does not. The ORACLE trial [17] is the cautionary precedent: a model that raised agreement without raising accuracy. Any deployment claim on this benchmark should therefore state the decision, the threshold, and the base-rate baseline it must beat — none of which a headline ROC number supplies.

Responsible disclosure. We frame every code-level defect in this paper (D1–D10, including the un-trained mode-experts mechanism, the empty eligibility-criteria cache, and the binarised PR-AUC computation) as an ecosystem reproducibility hazard, not as author misconduct. Each is reproducible from the unmodified public repositories, and several — the empty cache and the released-checkpoint reload variance in particular — silently affect anyone who re-runs the code, which is precisely why they matter. Following responsible-disclosure practice, we reported the defects to the respective project maintainers through their public issue trackers before submission: HINT issue #15 (<https://github.com/futianfan/clinical-trial-outcome-prediction/issues/15>; audited commit 8dc0497) and MEXA-CTP issue #2 (<https://github.com/murai-lab/MEXA-CTP/issues/2>; audited commit b898dbb), both also recorded in DISCLOSURE.md with the code release. Our aim is to harden a shared benchmark that the whole subfield depends on, and we would welcome corrections from the original authors.

Reproducibility. All code, raw per-trial predictions and result tables are released at <https://github.com/azizcetinerr/top-benchmark-reeval>. All defect audits are time-stamped: the HINT repository was audited at commit 8dc0497 and MEXA-CTP at commit b898dbb (both July 2026; full hashes in DISCLOSURE.md). Every table in this paper is regenerated from stored predictions without retraining. Settings: seed 2023, ICD-10-CM 2024, BioBERT dmis-lab/biobert-v1.1, 1000 bootstrap resamples at 95% for all confidence intervals and paired tests, except the phase-III multiple-comparison stress test (Section 4.2.7), which uses 10,000 resamples to stabilise the corrected p -values. The tabular baselines of Section 4.2.3 additionally use TabFM (tabfm-1.0.0-jax) with 32 ensemble members and RDKit 2026.03 for molecular descriptors and Morgan fingerprints. Feature construction: RDKit descriptors are aggregated across a trial’s drugs by mean and max; Morgan/ECFP fingerprints use radius 2 and 256 bits, combined across drugs by bit-union (a bit is set if present in any drug); GRAM disease features are multi-hot over ontology ancestors. The tuned gradient-boosting baselines (Section 4.2.2) use XGBoost 3.2, LightGBM 4.7 and CatBoost 1.2, each grid-selected on validation.

Data availability

All code, raw per-trial predictions and result tables generated in this study are publicly available at <https://github.com/azizcetinerr/top-benchmark-reeval>. The underlying TOP benchmark and the HINT and MEXA-CTP implementations are third-party resources, publicly available from their original repositories (audited at commits 8dc0497 and b898dbb); TOP itself is derived from de-identified, publicly accessible ClinicalTrials.gov records. No new data were generated.

Declaration of competing interests

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CRedit authorship contribution statement

Aziz Muammer Çetiner: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Kolawole John Adebayo:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to help identify candidate tools, methods, and metrics that could be used to measure and evaluate specific aspects of the analysis. All methodological choices were evaluated, selected, and verified by the authors, who reviewed and edited all content and take full responsibility for the content of the published article.

Ethics statement

This study did not require ethical approval. It is a secondary, computational re-analysis of the publicly available TOP benchmark, which is derived from de-identified, publicly accessible ClinicalTrials.gov records. No human participants, human data, or biological materials were collected or accessed by the authors, and no new clinical data were generated.

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